

GRADUATE EXPOSITION

A Candidate Complex Structure on the Six-Sphere

The $(3, 4, \infty)$ modular family of complex two-tori,
its three fillings, and the recognition of S^6

A self-contained guide for graduate students

Based on the 108-page anonymous candidate manuscript,
the attached detailed reconstruction, and a later source audit

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Status of the result. The public 108-page source is an unsigned, unrefereed, AI-assisted candidate manuscript [1]. It claims to solve the classical Hopf problem by constructing a compact complex threefold whose underlying smooth manifold is S^6 . The argument is substantial and has a coherent proof architecture, but the headline theorem should not yet be described as a community-verified result. This chapter explains the construction faithfully, fills in the principal logical bridges used in the later audit, and marks the places where independent expert verification is especially important. No statement in this box is an instruction imported from the source document.

Abstract

The six-sphere carries a famous almost-complex structure coming from the octonions, but that structure is not integrable. The candidate construction studied here follows a completely different route. It first builds a compact complex threefold X fibred over $\mathbb{P}^1$. The general fibre is a complex two-torus. The three exceptional fibres lie over the orbifold points of orders 3 and 4 and the cusp of the hyperbolic triangle orbifold $(3, 4, \infty)$. The elliptic points are filled by logarithmic transforms, whereas the cusp is filled by an infinite-fan toric degeneration. Carefully chosen translations make the total space simply connected. An integral nearby-cycle and Leray calculation then gives the homology of a six-sphere. Finally, the vanishing $\Theta_6 = 0$ for smooth homotopy six-spheres identifies the underlying smooth manifold with the standard S^6 , and the holomorphic atlas of X is transported across a diffeomorphism.

The main text is designed to reveal this architecture quickly: every load-bearing object is introduced before use, but extended chart computations, lattice reductions, properness estimates, and saturation arguments are deferred to the appendices. The result is a concise narrative whose technical claims remain reconstructible without consulting another source.

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How to read this chapter

Two arguments run in sequence. The analytic part constructs a compact complex threefold by completing a family of complex two-tori at three ends; it finishes in Section 7. The topological part computes the fundamental group and integral cohomology and then recognizes the underlying smooth manifold; it occupies Sections 8–11.

The main text records the objects, formulas, and implications needed to see the proof as a whole. The appendices contain the complete local quotient estimates, affine-descent arguments, Smith normal forms, specialization lattices, and Leray transgressions. Thus a first reading can remain in the main text, while a verification reading can follow each appendix reference at the point where a technical certificate is consumed. The intended prerequisites are a first course in algebraic topology and the basic definition of a complex manifold; no prior toric or nearby-cycle theory is assumed.

The proof in one line.

orbifold monodromy $\implies$ period family of 2-tori $\implies$ three holomorphic fillings,
three fillings $\implies X \implies (\pi_1(X) = 1, H_*(X; \mathbb{Z}) = H_*(S^6; \mathbb{Z})) \implies X \cong_{\text{diff}} S^6$.

The complex atlas comes first; the identification with the sphere comes last.

1 The Hopf problem and the meaning of the claim

1.1 Almost-complex versus complex

Let M be a smooth real manifold of even dimension $2n$.

Definition 1.1. An *almost-complex structure* on M is a smooth bundle endomorphism $J : TM \rightarrow TM$ satisfying $J^2 = -\text{id}$.

Pointwise, such a J makes every tangent space into a complex vector space. What it does *not* automatically provide is a system of local coordinates with holomorphic transition maps. The obstruction is the Nijenhuis tensor

$$N_J(X, Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y].$$

The Newlander–Nirenberg theorem states that a smooth almost-complex structure is induced by a complex atlas if and only if $N_J = 0$. In this case J is called *integrable* [4].

The unit sphere in the imaginary octonions carries a natural almost-complex structure: at $x \in S^6 \subset \text{Im } \mathbb{O}$, define $J_x(v) = xv$ on $T_x S^6 = x^\perp$. This is the standard nearly Kähler structure. It is not integrable. The Hopf problem asks whether *some other* almost-complex structure on the underlying smooth sphere is integrable; see [2] for a historical account and precise formulations of the problem.

Among positive-dimensional even spheres, topological obstructions eliminate all except S^2 and S^6 from even admitting an almost-complex structure [3, 2]. The first is the Riemann sphere. Thus S^6 is the unique unresolved sphere in the classical problem.

1.2 What a positive solution must look like

If a complex threefold X is diffeomorphic to S^6 , then

$$H^2(X; \mathbb{R}) = 0.$$

A compact Kähler manifold has a nonzero Kähler class in $H^2(X; \mathbb{R})$, so a complex six-sphere must be non-Kähler. This rules out the most familiar construction tools from projective and Kähler geometry. It also warns us that the Hodge decomposition and the degeneration of the Frölicher spectral sequence cannot be used casually.

The candidate proof avoids constructing an integrable tensor directly on the round sphere. Instead it uses the following logically cleaner strategy.

1. Construct a compact complex manifold X of complex dimension three.
2. Compute the smooth topology of X and prove that X is diffeomorphic to S^6 .
3. Pull the holomorphic coordinate charts of X back through a diffeomorphism $S^6 \rightarrow X$.

This last step is elementary but important. The proof need not write a closed formula for an almost-complex tensor on the round sphere. A complex atlas on a diffeomorphic manifold is enough.

2 Conventions and background used throughout

This section collects the minimum vocabulary needed to follow the construction. It is intentionally concrete: every item below will reappear as an input to a later proof step.

2.1 Complex tori from period lattices

A *complex two-torus* is obtained from an abstract lattice $\Lambda \simeq \mathbb{Z}^4$ and a homomorphism $\Omega : \Lambda \rightarrow \mathbb{C}^2$ whose real-linear extension $\Omega_{\mathbb{R}} : \Lambda \otimes_{\mathbb{Z}} \mathbb{R} \rightarrow \mathbb{C}^2$ is an isomorphism. Equivalently, $\Omega(\Lambda)$ is a discrete subgroup generated by four real-linearly independent vectors, and

$$F = \mathbb{C}^2 / \Omega(\Lambda).$$

We shall consistently use Λ for the abstract lattice and $\Omega(\Lambda)$ for its embedded image in $\mathbb{C}^2$. If the four generators are the columns of a 2×4 complex matrix Ω , then the quotient is compact precisely when the associated real-linear map

$$\Omega : \mathbb{R}^4 \longrightarrow \mathbb{C}^2 \simeq \mathbb{R}^4$$

is invertible. No positivity or polarization condition is imposed: the fibres are general complex tori, not assumed to be abelian varieties.

The period map identifies

$$H_1(F; \mathbb{Z}) \simeq \Lambda, \quad H^1(F; \mathbb{Z}) \simeq V := \text{Hom}(\Lambda, \mathbb{Z}).$$

The cohomology ring is the exterior algebra

$$H^q(F; \mathbb{Z}) \simeq \bigwedge^q V. \tag{1}$$

Thus a matrix acting on Λ acts by the inverse transpose on V and by exterior powers on higher cohomology. This elementary dictionary is the reason that both homological matrices A_j and cohomological matrices T_j appear below.

2.2 Families, monodromy, and local systems

Let $f^\circ : J \rightarrow B^\circ$ be a smooth proper family of such tori. Parallel transport around a loop in B° acts on $H_1(F; \mathbb{Z})$; this gives the *monodromy representation*

$$\rho : \pi_1(B^\circ) \longrightarrow \mathrm{GL}(\Lambda).$$

The groups $H^q(F; \mathbb{Z})$ consequently vary as local systems

$$L^q = R^q f_*^\circ \mathbb{Z}, \quad (L^q)_b \simeq \bigwedge^q V.$$

For a loop with monodromy T , the invariant lattice $\ker(T - I)$ consists of classes that return unchanged, while the coinvariant lattice $V/(T - I)V$ records classes modulo monodromy differences. Here “over $\mathbb{Q}$ ” means after tensoring over $\mathbb{Z}$ with $\mathbb{Q}$. For example, the inclusion $2\mathbb{Z} \subset \mathbb{Z}$ becomes

$$(2\mathbb{Z}) \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q},$$

but integrally the quotient is $\mathbb{Z}/2\mathbb{Z}$. More generally, a finite-index sublattice $L \subset M$ satisfies $L \otimes_{\mathbb{Z}} \mathbb{Q} = M \otimes_{\mathbb{Z}} \mathbb{Q}$, even though M/L may be a nonzero finite torsion group. Thus rational linear algebra forgets precisely the finite-index information needed below. This is why the later calculations use Smith normal form and explicitly check saturation.

2.3 The three kinds of special fibre

Two local models occur.

- A *multiple smooth fibre* mS is locally obtained after a cyclic base change $t = s^m$ and a free cyclic quotient combining the rotation of s with a translation of the torus. The reduced surface is S , but the scheme-theoretic fibre has multiplicity m . This is the logarithmic transform used at the points of orders 3 and 4; the terminology and the classification of the resulting bielliptic surfaces are standard in complex surface theory [9].
- A *reduced normal-crossing fibre* is locally given by $z_0 \cdots z_{r-1} = 0$. In a one-parameter smoothing $z_0 \cdots z_{r-1} = t$, the local Milnor fibre retracts onto $(S^1)^{r-1}$. The cusp model is constructed from a toric fan, following the general circle of ideas of toroidal degenerations [7, 8]; all properness and freeness assertions needed here are nevertheless proved directly.

2.4 Two cohomological tools

We shall use two spectral sequences, always with integral coefficients. For a degeneration over a disc, the nearby-cycle sheaves $\mathcal{H}^q \psi$ have stalks equal to the cohomology of the local Milnor fibres. Their hypercohomology reconstructs the cohomology of a nearby smooth fibre. A standard reference for nearby and vanishing cycles is [10, Chapter 4]; Sections 9 and K compute the particular sheaves from scratch.

For the global map $f : X \rightarrow B$, the Leray spectral sequence is

$$E_2^{a,b} = H^a(B, R^b f_* \mathbb{Z}) \implies H^{a+b}(X; \mathbb{Z}). \quad (2)$$

Since $B \simeq S^2$, only the columns $a = 0, 1, 2$ can occur. Consequently the only possible nonzero differentials are $d_2 : E_2^{0,b} \rightarrow E_2^{2,b-1}$. The proof will identify their integral generators and compute all three coefficients, rather than infer them from rational ranks.

3 The global blueprint

Let

$$B = \mathbb{P}_t^1, \quad p_1 = (t = 0), \quad p_2 = (t = 1), \quad p_0 = (t = \infty),$$

and let $B^\circ = B \setminus \{p_0, p_1, p_2\}$. The construction produces a proper holomorphic map

$$f : X \longrightarrow B$$

with the following fibres.

Point of the base	Fibre	Local mechanism
$t \in B^\circ$	complex two-torus	period lattice in $\mathbb{C}^2$
p_1	$3S_1, S_1$ bielliptic	order-3 logarithmic transform
p_2	$4S_2, S_2$ bielliptic	order-4 logarithmic transform
p_0	reduced normal-crossing surface W	infinite-fan toric filling

The two smooth surface types in the table are not synonyms. A complex two-torus is a quotient $F = \mathbb{C}^2/\Omega(\Lambda)$; it is itself a compact complex Lie group, and it need not split as a product of elliptic curves. A *bielliptic surface*, by contrast, has the form

$$S = (E \times E')/G,$$

where E, E' are elliptic curves and the finite group G acts freely but not solely by translations. Thus $E \times E' \rightarrow S$ is a finite unramified cover by a complex two-torus, but S is not itself a complex torus. For example, their first Betti numbers distinguish them:

$$b_1(F) = 4, \quad b_1(S) = 2.$$

Equivalently on the holomorphic side, K_F is trivial, whereas K_S is a nontrivial torsion line bundle. These facts are part of the standard classification of compact complex surfaces [9].

The notation $3S_1$ and $4S_2$ records multiplicity, not the number of components. The scheme-theoretic fibre over p_1 has multiplicity three along the smooth reduced surface S_1 ; similarly, the fibre over p_2 has multiplicity four along S_2 . The reduced surfaces S_1, S_2 are bielliptic, whereas every fibre over B° is a complex two-torus.

3.1 The geometry of the base orbifold

The signature $(0; 3, 4, \infty)$ has a precise geometric meaning. The underlying compact Riemann surface has genus zero, hence is the ordinary sphere $B \simeq \mathbb{P}^1$. The points p_1 and p_2 carry cyclic stabilizers of orders 3 and 4, while p_0 is a cusp. Near a cone point of order m , a disc coordinate s is acted on by $s \mapsto e^{-2\pi i/m} s$, and the coarse quotient coordinate is $u = s^m$. Thus the coarse sphere is smooth at p_1, p_2 , although the orbifold retains cone angles $2\pi/3$ and $\pi/2$. The cusp instead comes from a *horodisc*. After moving its ideal centre to ∞ in the upper-half-plane model, such a region is

$$\mathcal{H}_Y := \{w \in \mathfrak{h} : \operatorname{Im} w > Y\}, \quad Y > 0. \quad (3)$$

The line $\operatorname{Im} w = Y$ is its bounding *horocycle*. Under a Cayley map to the unit disc, $\mathcal{H}_Y$ becomes a Euclidean disc internally tangent to the boundary; hence the name.

In complex dimension one, *horoball* is a synonym. Its quotient by an infinite-order parabolic translation is a punctured disc, as we now make explicit.

Its orbifold Euler characteristic is

$$\chi_{\text{orb}} = 2 - \left(1 - \frac{1}{3}\right) - \left(1 - \frac{1}{4}\right) - 1 = -\frac{5}{12} < 0,$$

so the orbifold is hyperbolic. Equivalently, the upper half-plane is tiled by a geodesic triangle with angles $\pi/3, \pi/4, 0$. The two finite vertices give clockwise rotations g_1, g_2 of orders 3, 4, and the ideal vertex gives the parabolic element $g_0 = (g_1 g_2)^{-1}$. Hence the orientation-preserving orbifold group is

$$\Gamma_{\text{orb}} = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle, \quad g_0 = (g_1 g_2)^{-1}, \quad (4)$$

and $\mathfrak{h}/\Gamma_{\text{orb}}$ has underlying Riemann surface $B \setminus \{p_0\}$. After conjugating the cusp so that $g_0(w) = w - 1$, each sufficiently high horodisc $\mathcal{H}_Y$ is g_0 -invariant. The function

$$q = e^{2\pi i w}$$

is invariant under $w \mapsto w - 1$ and induces a biholomorphism

$$\mathcal{H}_Y / \langle g_0 \rangle \simeq \{0 < |q| < e^{-2\pi Y}\}.$$

Thus the horodisc is the simply connected cusp neighbourhood upstairs, whereas its quotient is the punctured-disc neighbourhood downstairs; adjoining $q = 0$ fills the cusp. The construction uses only the power coordinates at the cone points, the stabilizer orders, and this parabolic cusp coordinate. Appendix A develops the triangle geometry and quotient coordinates in detail; see also [5].

Returning to the global construction, each of the three fillings contains an affine translation in a fibre direction. Most of that translation is erased by monodromy, and only its value under the invariant functional γ introduced below survives in the global topology. At a finite point the actual translation is v_j/m_j , where $(m_1, m_2) = (3, 4)$, and we record its integral numerator $\ell_j := \gamma(v_j)$; at the cusp, ℓ_0 records a possible integral regluing. Thus the three analytically rather different fillings affect the final topology through the single triple (ℓ_0, ℓ_1, ℓ_2) . For the chosen fillings it is

$$(\ell_0, \ell_1, \ell_2) = (0, 1, -1).$$

They enter both the fundamental group and the Leray differential through the same integer

$$p = 12\ell_0 - 4\ell_1 - 3\ell_2 = -1. \quad (5)$$

The fact that p is a unit in $\mathbb{Z}$ is the topological reason that all middle-dimensional homology disappears.

Outputs required from the construction. The analytic half must produce a Hausdorff compact complex threefold and a proper map $f : X \rightarrow \mathbb{P}^1$. The topological half must prove, integrally,

$$\pi_1(X) = 1, \quad H^k(X; \mathbb{Z}) = 0 \quad (1 \leq k \leq 5).$$

The first assertion comes from van Kampen and the second from the Leray spectral sequence. Neither calculation can be replaced by a rational one, because finite-index errors would survive rationalization.

4 The integral monodromy data

4.1 The design problem

The two matrices below are the algebraic hinge of the construction, and it is useful to understand the problem they were designed to solve before reading their entries. The monodromy must satisfy several analytic and topological requirements simultaneously:

- (i) the monodromies at the two cone points must have orders 3 and 4;
- (ii) the inverse product must be unipotent with $(T_0 - I)^2 = 0$, with a primitive toric image and a unimodular quotient-to-image map;
- (iii) a common primitive invariant γ must detect the only global coinvariant circle, and later fibre translations must kill that circle;
- (iv) it must admit the required holomorphic period realization.

A practical way to search for such matrices is to begin with the familiar rank-two integral rotations

$$r_3 : (u, w) \mapsto (-u - w, u), \quad r_4 : (u, w) \mapsto (w, -u),$$

where the ordered pair records the images of the basis vectors. The formula for r_3 is the usual rotation through 120 degrees, but written in an oblique integral basis of the hexagonal lattice rather than in an orthonormal basis. To see this, realize the two basis vectors in the Euclidean plane as

$$u = (1, 0), \quad w = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

They have equal length and meet at an angle of 120 degrees, and

$$-u - w = \left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right).$$

Thus the three rays generated by $u, -u - w, w$ are separated successively by 120 degrees, and

$$u \xrightarrow{r_3} -u - w \xrightarrow{r_3} w \xrightarrow{r_3} u.$$

The same cycle contains both basis vectors, so $r_3^3 = I$; since $r_3(u) \neq u$, its order is exactly 3. The nonstandard-looking integral formula is precisely what the ordinary rotation matrix becomes in this hexagonal-lattice basis. Likewise,

$$(u, w) \xrightarrow{r_4} (w, -u) \xrightarrow{r_4} (-u, -w) \xrightarrow{r_4} (-w, u) \xrightarrow{r_4} (u, w),$$

so r_4 has order exactly 4. One then adjoins the invariant direction γ and an extension direction δ , and solves integral linear conditions requiring the inverse product to have the prescribed primitive nilpotent part. In this sense the entries 6, 2, and -3 below are extension coefficients produced by simultaneous constraints; they are not independent guesses. Their reappearance in the period equations is one of the internal consistency checks of the construction.

These requirements do not assert that the answer is unique. Rather, the displayed matrices are one explicit integral solution to this design problem. The calculations in the rest of this section are therefore certificates, not decorations: each one verifies a feature that will later be consumed by the period construction, the three fillings, or the topology of the total space.

4.2 Two dual lattices

Let

$$\Lambda = \mathbb{Z}\langle \hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta} \rangle, \quad V = \text{Hom}(\Lambda, \mathbb{Z}) = \mathbb{Z}\langle \gamma, u, w, \delta \rangle. \quad (6a)$$

It is convenient to display the action first on V : the later Leray calculation uses the exterior powers $\Lambda^q V$, whereas the fibre translations themselves live in Λ . Duality then recovers the latter action by inverse transpose, so the two calculations remain synchronized. The lattice Λ will be H_1 of a smooth torus fibre, while V will be its H^1 , in accordance with (1). In the basis of V , the following integral matrices realize the design constraints above:

$$T_1 = \begin{pmatrix} 1 & 0 & -6 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad T_2 = \begin{pmatrix} 1 & 6 & 0 & -3 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (6b)$$

The monodromy around the third orbifold loop is determined by the relation $g_0 g_1 g_2 = 1$. Accordingly, on V set

$$T_0 := (T_1 T_2)^{-1}.$$

Direct multiplication then gives

$$T_1^3 = T_2^4 = I, \quad T_0 = I + N, \quad N = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad N^2 = 0.$$

Thus the first two monodromies have finite orders 3 and 4, while the cusp monodromy is unipotent of index two.

We now pass from cohomology to homology. Since $V = \text{Hom}(\Lambda, \mathbb{Z})$ and the displayed bases are dual, define, for $j = 0, 1, 2$,

$$A_j := (T_j^{-1})^t.$$

Because $A_1^3 = A_2^4 = I$, the assignments $g_1 \mapsto A_1$ and $g_2 \mapsto A_2$ extend uniquely through the presentation (4) to the homological monodromy representation

$$A : \Gamma_{\text{orb}} \longrightarrow \text{GL}(\Lambda).$$

For $g \in \Gamma_{\text{orb}}$, the notation $A(g)$ means its image under this representation. In particular,

$$A(g_0) = A_0 = (T_0^{-1})^t = (A_1 A_2)^{-1}.$$

Writing this homological cusp action in the basis of Λ gives

$$A_0 \hat{\gamma} = \hat{\gamma} - \hat{\delta}, \quad A_0 \hat{u} = \hat{u} + \hat{w}, \quad A_0 \hat{w} = \hat{w}, \quad A_0 \hat{\delta} = \hat{\delta}.$$

Consequently

$$\Lambda_{\text{tor}} = \ker(A_0 - I) = \text{im}(A_0 - I) = \mathbb{Z}\langle \hat{w}, \hat{\delta} \rangle. \quad (7)$$

We call this the *toric sublattice* of the cusp. The subscript tor is a mnemonic for its toric, or toroidal, role below; it does *not* mean “torsion.” Indeed, the displayed basis shows that Λ_{tor} is a primitive rank-two direct summand of Λ , so $\Lambda/\Lambda_{\text{tor}} \simeq \mathbb{Z}^2$ is torsion free. The name anticipates the cusp construction: quotienting $\mathbb{C}^2$ by the periods in Λ_{tor} and then exponentiating produces the algebraic torus $(\mathbb{C}^*)^2$. The induced map

$$B_0 : \Lambda/\Lambda_{\text{tor}} \longrightarrow \Lambda_{\text{tor}}, \quad B_0(\bar{\hat{\gamma}}) = -\hat{\delta}, \quad B_0(\bar{\hat{u}}) = \hat{w},$$

has matrix

$$[B_0] = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

In particular, B_0 is unimodular. This one fact is used twice: it makes the toric quotient free near the cusp, and it makes the quotient of the infinite fan have only one component orbit.

4.3 The surviving circle and the twist vectors

The functional $\gamma \in V$ is fixed by both T_1 and T_2 . Equivalently, $\gamma \circ A_j = \gamma$. Let

$$L = \ker \gamma = \mathbb{Z}\langle \hat{u}, \hat{w}, \hat{\delta} \rangle.$$

The sum of the images $(A_1 - I)\Lambda + (A_2 - I)\Lambda$ is exactly this saturated rank-three lattice. Hence the global coinvariants of Λ are detected by the single quotient

$$\Lambda/L \simeq \mathbb{Z},$$

with coordinate γ . Informally, three fibre directions are destined to be killed by monodromy and the cusp; one circle remains, and the logarithmic twists must kill it.

This turns the choice of translations into a small integral design problem. Over the cone point of order m_j , where $(m_1, m_2) = (3, 4)$, we want a primitive vector v_j fixed by A_j : then m_j repetitions of translation by v_j/m_j add to the integral period v_j . To make the resulting affine action free, the surviving character provides a simple criterion: we require $\gamma(v_j)$ to be relatively prime to m_j . Finally, to make the topological integer in (5) a unit, we choose the signed values

$$\ell_1 = \gamma(v_1) = 1, \quad \ell_2 = \gamma(v_2) = -1.$$

Solving $(A_j - I)v_j = 0$ with these character values gives the convenient primitive solutions

$$\varepsilon = \hat{\gamma} + 2\hat{u} - 4\hat{w}, \quad \varepsilon' = \hat{\gamma} + 3\hat{u} - 3\hat{w}, \quad v_1 = \varepsilon, \quad v_2 = -\varepsilon'. \quad (8)$$

Then

$$A_1 v_1 = v_1, \quad A_2 v_2 = v_2, \quad \gamma(v_1) = 1, \quad \gamma(v_2) = -1.$$

The fixed-vector identities are the analytic input that makes the fractional translations close after m_j turns; the two character values are the topological input that kills the final coinvariant circle. The same vectors therefore do two jobs, which is the main reason for choosing them at this early stage.

5 The period map and the smooth torus family

The integral representation constructed above gives the topology of a torus bundle, but it does not yet put complex structures on its fibres. We must embed the abstract lattice Λ as a holomorphically varying full lattice in $\mathbb{C}^2$, and the embedding must intertwine the monodromy action. There are three logically separate tasks. First construct a modular parameter τ with the correct branching at the order-3 and order-4 points. Next let period-lattice descent force the two affine functions μ and β . Finally use the additive freedom in β to make the four period vectors real-linearly independent. This triangular order is what makes an apparently nonlinear period problem manageable.

5.1 The classical modular invariant j

The first period function is obtained by inverting the modular invariant, so we recall precisely which normalization and which local branching data are being used. Let

$$\mathfrak{h}_\tau = \{\tau \in \mathbb{C} : \text{Im } \tau > 0\},$$

on which $\text{PSL}_2(\mathbb{Z})$ acts by fractional linear transformations. We shall use the standard generators

$$S(\tau) = -\frac{1}{\tau}, \quad T(\tau) = \tau + 1.$$

The point τ determines the elliptic curve

$$E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}),$$

and two such curves are biholomorphic exactly when their parameters lie in the same $\text{PSL}_2(\mathbb{Z})$ -orbit. Thus $\text{PSL}_2(\mathbb{Z}) \backslash \mathfrak{h}_\tau$ is the coarse moduli space of complex elliptic curves.

The *classical modular invariant* is the $\text{PSL}_2(\mathbb{Z})$ -invariant holomorphic function

$$j(\tau) = \frac{E_4(\tau)^3}{\Delta(\tau)} = q^{-1} + 744 + 196884q + O(q^2). \quad (9)$$

Here $q = e^{2\pi i \tau}$ and E_4, Δ are the normalized Eisenstein and discriminant forms. Their precise Fourier series and transformation laws are recalled in Appendix B. What matters in the main construction is that j classifies elliptic curves:

$$\text{PSL}_2(\mathbb{Z}) \backslash \mathfrak{h}_\tau \simeq \mathbb{C}_j.$$

After adjoining the single cusp orbit, represented by ∞ , this becomes $\text{PSL}_2(\mathbb{Z}) \backslash \mathfrak{h}_\tau^* \simeq \mathbb{P}_j^1$. Some authors write $J = j/1728$; throughout this paper, j means the classically normalized function in (9).

There are two exceptional orbits in the upper half-plane. Let

$$\rho = e^{2\pi i/3} = \frac{-1 + i\sqrt{3}}{2}.$$

The stabilizers of ρ and i in $\text{PSL}_2(\mathbb{Z})$ have orders 3 and 2. They represent the hexagonal and square lattices, whose elliptic curves have extra automorphisms. Consequently j has the local expansions

$$j(\tau) = a_3(\tau - \rho)^3 + O((\tau - \rho)^4), \quad j(\tau) - 1728 = a_2(\tau - i)^2 + O((\tau - i)^3). \quad (10)$$

for nonzero constants a_3, a_2 . Thus $j(\rho) = 0$, $j(i) = 1728$, its ramification indices there are 3 and 2, and its q -expansion gives a simple pole at the cusp. Since $j(\tau + 1) = j(\tau)$, the same cubic model holds at $\rho + 1 = e^{\pi i/3}$. On the coarse quotient j is a coordinate; the ramification belongs to the invariant map from $\mathfrak{h}_\tau$ before taking the quotient. Appendix B derives these facts from E_4, E_6, Δ ; see also [6, Chapters 1–3].

5.2 Uniformizing the orbifold and constructing τ

Recall the presentation (4): g_1 and g_2 are the clockwise rotations of orders 3 and 4, while $g_0 = (g_1 g_2)^{-1}$ is the parabolic generator at the cusp. Recall also from Section 3 that t is the affine coordinate on $B = \mathbb{P}_t^1$, with

$$p_1 = (t = 0), \quad p_2 = (t = 1), \quad p_0 = (t = \infty).$$

Let $\mathfrak{h}_z$ be an upper half-plane on which Γ_{orb} acts, and denote the quotient map by

$$\pi : \mathfrak{h}_z \longrightarrow \mathfrak{h}_z / \Gamma_{\text{orb}} = B \setminus \{p_0\}.$$

The map π has ramification orders 3 and 4 over p_1 and p_2 . For $j = 1, 2$, choose a fixed point z_j of g_j above p_j and a local coordinate s_j so that

$$s_j(g_j z) = e^{-2\pi i / m_j} s_j(z), \quad m_1 = 3, \quad m_2 = 4.$$

Both rotations are clockwise. Their orientations cannot be reversed independently: a mixed choice would change the product from the parabolic cusp class to a hyperbolic element.

The choice of the factor 1728 is now transparent. On the compactified base, the function $1728t$ matches the three distinguished values of the classical modular invariant:

point of B	p_1	p_2	p_0
$1728t$	0	1728	∞
corresponding value of j	$j(\rho) = 0$	$j(i) = 1728$	$j(\infty) = \infty$

Moreover, since t is a local coordinate at p_1 and $t-1$ is a local coordinate at p_2 , the ramification of π gives

$$\text{ord}_{z_1}(t \circ \pi) = 3, \quad \text{ord}_{z_2}((t-1) \circ \pi) = 4.$$

Our aim is therefore to construct a holomorphic map $\tau : \mathfrak{h}_z \rightarrow \mathfrak{h}_\tau$ such that

$$j \circ \tau = 1728t \circ \pi.$$

We use the standard fractional-linear action

$$M \cdot \tau = \frac{a\tau + b}{c\tau + d}, \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z}), \quad \tau \in \mathfrak{h}_\tau.$$

The central matrices I and $-I$ induce the same transformation, so the effective acting group is $\text{PSL}_2(\mathbb{Z})$.

The desired equivariance of τ is not an independent choice. Recall from (6a) that u, w are the middle two vectors of the chosen basis (γ, u, w, δ) of the cohomology lattice $V = \text{Hom}(\Lambda, \mathbb{Z})$. On the rank-two sublattice $\mathbb{Z}\langle u, w \rangle$, the rotations used to design the full monodromy matrices T_1, T_2 in (6b) have matrices

$$[r_3]_{(u,w)} = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \quad [r_4]_{(u,w)} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Since an overall sign is invisible in the fractional-linear action, their projective classes may be represented by

$$M_1 = -[r_3]_{(u,w)} = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \quad M_2 = [r_4]_{(u,w)} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Thus $M_1 \cdot \tau = (\tau - 1)/\tau$ and $M_2 \cdot \tau = -1/\tau$. Moreover,

$$M_1 M_2 = - \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

so the inverse product acts by $\tau \mapsto \tau - 1$. Formula (11) below is therefore the modular shadow of the same two rank-two rotations from which T_1 and T_2 were built.

The local orders are exactly compatible with the branching of j described in (10): order 3 over p_1 cancels the cubic branch over $j = 0$, while order 4 over p_2 is divisible by the quadratic branch over $j = 1728$. The latter divisibility matches the preceding algebra: g_2 has order 4, whereas $M_2^2 = -I$, so the class of M_2 has order 2 in $\mathrm{PSL}_2(\mathbb{Z})$.

These branching compatibilities eliminate the local monodromy of an inverse of j . Consequently, analytic continuation of a local inverse produces the desired holomorphic map τ . Indeed, after removing the inverse images of p_1, p_2 , a small meridian over p_1 becomes a cube around the order-three branch of j , and one over p_2 becomes a fourth power around the order-two branch. Hence the local inverse has no monodromy around either deleted orbit. Local extraction of a cube root and a square root, respectively, extends τ holomorphically across those orbits. Appendix F supplies the global continuation argument and shows that one *simultaneous* modular conjugation realizes the two target matrices M_1, M_2 . Consequently,

$$\tau \circ g_1 = \frac{\tau - 1}{\tau}, \quad \tau \circ g_2 = -\frac{1}{\tau}, \quad \tau \circ g_0 = \tau - 1. \quad (11)$$

The first transformation fixes $\rho + 1 = e^{\pi i/3}$, which is $\mathrm{PSL}_2(\mathbb{Z})$ -equivalent to ρ ; the second fixes i ; and the third is a primitive parabolic translation at the cusp. Thus all three formulas are compatible with the special values and local models described in Section 5.1.

5.3 The two affine period functions

We now explain where the transformation laws for the remaining two period functions come from. They are not guessed: they are forced by the requirement that the period lattice descend under the monodromy matrices in (6b).

Start provisionally with four holomorphic functions a, b, c, d on $\mathfrak{h}_z$ and the normalized period matrix

$$\Omega_Z(z) = [Z(z) \mid I_2], \quad Z(z) = \begin{pmatrix} a(z) & b(z) \\ c(z) & d(z) \end{pmatrix}.$$

The last two columns have been normalized to the standard basis of $\mathbb{C}^2$. This is merely a choice of complex coordinates on the universal cover of each torus fibre.

We first derive the only coordinate changes that could make this provisional matrix descend. For a fixed Z , call $C_j^Z(z) \in \mathrm{GL}_2(\mathbb{C})$ a *provisional descent matrix* if

$$\Omega_Z(g_j z) = C_j^Z(z) \Omega_Z(z) A_j^{-1} = C_j^Z(z) \Omega_Z(z) T_j^t, \quad j = 1, 2. \quad (12)$$

Here $A_j = (T_j^{-1})^t$ is the action on the homology lattice, as defined above, whereas left multiplication by C_j^Z changes the complex coordinates on $\mathbb{C}^2$. At this stage Z is still unknown, and C_j^Z is only notation for a matrix associated with this trial Z ; the final matrices R_{g_j} will be defined only after Z has been constructed.

The normalization by I_2 determines C_j^Z before it imposes any transformation law on a, b, c, d . Let Q_j^Z be the rightmost 2×2 block of $\Omega_Z T_j^t$. Direct multiplication using (6b) gives

$$Q_1^Z = \begin{pmatrix} -b & 0 \\ 1-d & 1 \end{pmatrix}, \quad Q_2^Z = \begin{pmatrix} b & 0 \\ d & 1 \end{pmatrix}.$$

The rightmost block of $\Omega_Z(g_j z)$ is I_2 , so (12) forces $C_j^Z Q_j^Z = I_2$. Hence, whenever $b \neq 0$, there is exactly one possible choice:

$$C_1^Z = (Q_1^Z)^{-1} = \begin{pmatrix} -1/b & 0 \\ (1-d)/b & 1 \end{pmatrix}, \quad C_2^Z = (Q_2^Z)^{-1} = \begin{pmatrix} 1/b & 0 \\ -d/b & 1 \end{pmatrix}. \quad (13)$$

Working on the locus $b \neq 0$ causes no restriction in the eventual construction: below we set $b = \tau$, and $\tau(\mathfrak{h}_z) \subset \mathfrak{h}_\tau$ implies $\tau(z) \neq 0$ for every z .

The two rightmost columns have now done their entire job: they uniquely determine the provisional coordinate changes (13). The first two columns perform a different job: they decide which trial matrices Z actually descend. Substituting (13) into (12) and comparing the left 2×2 blocks yields the following formulas. Every uncomposed entry a, b, c, d on the right-hand side is evaluated at z .

$$Z(g_1 z) = \begin{pmatrix} (6-a)/b & (b-1)/b \\ c+2+(1-d)(a-6)/b & (1-d)/b \end{pmatrix}, \quad Z(g_2 z) = \begin{pmatrix} 6+a/b & -1/b \\ c-3-ad/b & 1+d/b \end{pmatrix}. \quad (14)$$

For instance, the first row of the first matrix says $a \circ g_1 = (6-a)/b$ and $b \circ g_1 = (b-1)/b$. Thus (14) is both necessary and sufficient for the two provisional matrices in (13) to satisfy the full descent identity (12).

These formulas also explain the otherwise mysterious coefficient 6. If one looks for a slice of the simple form $a = \kappa d$ that is preserved by both generators, the transformation laws for a and d force $\kappa = 6$. Indeed, g_1 -invariance requires $(6 - \kappa d)/b = \kappa(1-d)/b$, and hence $\kappa = 6$; the g_2 -laws give the same conclusion. We therefore restrict to the invariant slice

$$b = \tau, \quad d = \mu, \quad a = 6\mu, \quad c = \beta.$$

The laws for b in (14) are precisely the first two laws for τ in (11). The laws for d and c now force us to seek holomorphic functions $\mu, \beta : \mathfrak{h}_z \rightarrow \mathbb{C}$ satisfying

$$\mu \circ g_1 = \frac{1-\mu}{\tau}, \quad \mu \circ g_2 = 1 + \frac{\mu}{\tau}, \quad (15)$$

$$\beta \circ g_1 = \beta + 2 - \frac{6(1-\mu)^2}{\tau}, \quad \beta \circ g_2 = \beta - 3 - \frac{6\mu^2}{\tau}. \quad (16)$$

Thus the quadratic terms arise by substituting $a = 6d$ into the c -laws, while the constants 2 and -3 are the corresponding integral entries of T_1 and T_2 in (6b). This is the geometric motivation for all four affine equations. These are affine descent equations. Local solutions exist at the two finite stabilizers by averaging and at the cusp by direct substitution. The differences of two local solutions form, respectively, the line-bundle sheaves

$$\mathcal{O}_{\mathbb{P}^1}(-1) \quad \text{for } \mu, \quad \mathcal{O}_{\mathbb{P}^1} \quad \text{for } \beta.$$

Since both first cohomology groups vanish, the local solutions glue. Thus μ exists uniquely and β exists uniquely up to an additive constant. Appendix F derives the two sheaves from the modular multiplier, checks its signs and orders, and gives the local solutions explicitly.

5.4 Nondegeneracy of the period lattice

The additive freedom in β is used to arrange

$$D := \operatorname{Im} \beta - \frac{6(\operatorname{Im} \mu)^2}{\operatorname{Im} \tau} < 0 \quad \text{on } \mathfrak{h}_z. \quad (17)$$

The transformation laws make D invariant. Near the cusp, $\operatorname{Im} \tau \rightarrow +\infty$ while μ and $\beta + \tau$ remain bounded, so $D \rightarrow -\infty$. On the compact remainder of the quotient it is bounded above; shifting β by a constant with sufficiently negative imaginary part therefore gives $D < 0$ everywhere.

Now define the period matrix

$$\Omega(z) = \begin{pmatrix} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{pmatrix} = [Z(z) \mid I_2]. \quad (18)$$

From this point onward, Z denotes this specialized left block $\begin{pmatrix} 6\mu & \tau \\ \beta & \mu \end{pmatrix}$, not the earlier four-function trial matrix. Regarded as a real-linear map $\Lambda \otimes \mathbb{R} \rightarrow \mathbb{C}^2$, this matrix has determinant

$$\det_{\mathbb{R}} \Omega = -\det(\operatorname{Im} Z) = \operatorname{Im} \tau D \neq 0. \quad (19)$$

The sign and the real block-matrix calculation are written out in Appendix C. Since $\operatorname{Im} \tau > 0$ and $D < 0$, the four columns of Ω are real-linearly independent. Therefore $\Omega(z)\Lambda$ is a full rank-four lattice in $\mathbb{C}^2$, and

$$F_z = \mathbb{C}^2 / \Omega(z)\Lambda$$

is a complex two-torus.

We can now pass from the provisional matrices C_j^Z to the actual coordinate changes for this torus family. Define, once and for all on the two generators,

$$R_{g_j}(z) := C_j^Z(z) \Big|_{b=\tau, d=\mu}, \quad j = 1, 2.$$

Explicitly, formula (13) gives

$$R_{g_1}(z) = \begin{pmatrix} -1/\tau(z) & 0 \\ (1 - \mu(z))/\tau(z) & 1 \end{pmatrix}, \quad R_{g_2}(z) = \begin{pmatrix} 1/\tau(z) & 0 \\ -\mu(z)/\tau(z) & 1 \end{pmatrix}.$$

This is the definitive meaning of the symbols R_{g_1} and R_{g_2} for the final family. Every later R_g will be a cocycle product of these generator matrices, not a new choice. They are holomorphic and invertible because $\tau(z) \in \mathfrak{h}_\tau$, hence $\tau(z) \neq 0$.

The functions τ, μ, β satisfy (11), (15), and (16); together with $a = 6\mu$, $b = \tau$, $c = \beta$, and $d = \mu$, these are exactly the specialization of (14). Therefore the provisional identity (12) specializes to the generator equivariance identities

$$\Omega(g_j z) = R_{g_j}(z) \Omega(z) A_j^{-1}, \quad j = 1, 2. \quad (20)$$

Only now do we define R_g for an arbitrary group element. Put $R_e = I_2$. For a word $g = h_1 \cdots h_r$ in g_1, g_2 —inverses may be replaced by positive powers using $g_1^{-1} = g_1^2$ and $g_2^{-1} = g_2^3$ —set

$$R_g(z) := R_{h_1}(h_2 \cdots h_r z) R_{h_2}(h_3 \cdots h_r z) \cdots R_{h_r}(z).$$

Equivalently, this definition is generated recursively by the cocycle rule

$$R_{gh}(z) = R_g(hz) R_h(z).$$

It remains to show that the resulting matrix depends only on the group element, not on the chosen word. By (4), it is enough to check the two relation words g_1^3 and g_2^4 . Iterating (20) m_j times gives

$$\Omega(z) = R_{g_j}^{m_j}(z) \Omega(z) A_j^{-m_j} = R_{g_j}^{m_j}(z) \Omega(z), \quad (m_1, m_2) = (3, 4).$$

Because $\Omega(z) : \Lambda \otimes \mathbb{R} \rightarrow \mathbb{C}^2$ is a real-linear isomorphism, this forces $R_{g_j}^{m_j}(z) = I_2$. Hence both defining relations act trivially, and R_g is well defined for every $g \in \Gamma_{\text{orb}}$.

Combining the generator identities with the cocycle rule gives

$$\Omega(gz) = R_g(z)\Omega(z)A(g)^{-1}, \quad R_{gh}(z) = R_g(hz)R_h(z). \quad (21)$$

Here $A(g)$ acts on the abstract lattice, whereas $R_g(z)$ acts complex linearly on $\mathbb{C}^2$. The displayed identity says precisely that the two actions carry the embedded period lattice over z to the embedded period lattice over gz . Taking the quotient of $(\mathfrak{h}_z \times \mathbb{C}^2)/\Lambda$ by the free part of the orbifold action gives a proper holomorphic family

$$J \longrightarrow B^\circ$$

of complex two-tori. Locally on the base it is the quotient of $U \times \mathbb{C}^2$ by a continuously varying full lattice. Hence it is a smooth torus bundle with compact fibres, and the projection is proper over B° .

6 The three holomorphic fillings

The family $J \rightarrow B^\circ$ is already complete over the pair of pants B° , but it has three analytically different ends. At p_1 and p_2 the monodromy has finite order. There one passes to a cyclic cover of the base, fills in a smooth torus fibre, and divides by a carefully chosen free affine action. At the cusp p_0 , by contrast, the monodromy is unipotent of infinite order. The missing fibre must then be supplied by a semistable toric model built from an infinite periodic fan.

For each filling we must verify four points, not merely write down a formal quotient: the local total space is a Hausdorff complex manifold; its map to the disc is proper; the special fibre has the stated structure; and over the punctured disc the new model is biholomorphic to the family constructed in Section 5. We carry out these checks in that order below.

6.1 The cusp and its infinite fan

The cusp cannot be treated by the same finite cyclic quotient used at the two cone points: its monodromy is unipotent of infinite order, so no finite base change makes the family constant. The useful observation is that the period matrix has only a logarithmic singularity. The strategy is therefore to divide first by the two invariant periods, turning the additive fibre coordinates into $(\mathbb{C}^*)^2$; exponentiation then turns the logarithmic term into integral powers of the cusp coordinate; and toric geometry is precisely the framework in which such monomials extend across that coordinate's zero. The infinite fan below is the equivariant compactification required by the remaining rank-two lattice action. Appendix G gives a self-contained account of the toric notions used in this step, from cones and fans to the local equation $t_c = z_0 z_1 z_2$ and the periodic fan quotient.

From additive periods to multiplicative coordinates

Put $t_c = 1/t$, a local coordinate at $p_0 = (t = \infty)$, and use the same symbol for its pullback $t_c \circ \pi$ to $\mathfrak{h}_z$. Recall the horodisc $\mathcal{H}_Y$ from (3). Here it is understood after the conjugation that places the chosen lift of the cusp at ∞ and makes g_0 act by $w \mapsto w - 1$. Making the cusp neighbourhood “sufficiently small” downstairs means taking Y sufficiently large upstairs: indeed, its quotient has radius $e^{-2\pi Y}$. The distinction matters for the next step. The punctured disc downstairs is not simply connected, and its coordinate t_c has no single-valued logarithm there. The horodisc $\mathcal{H}_Y$ upstairs, however, is simply connected, and $t_c \circ \pi$ is holomorphic

and nowhere zero on it. It therefore has a holomorphic logarithm on $\mathcal{H}_Y$. We normalize that logarithm by writing

$$e^{2\pi i s} = t_c, \quad s \circ g_0 = s - 1.$$

The minus sign records orientation. Since $t_c \circ \pi$ is g_0 -invariant, $s(g_0 z) - s(z)$ is an integer, namely the signed winding number of the deck transformation. Our convention $g_0 = (g_1 g_2)^{-1}$ and the normalization $\tau \circ g_0 = \tau - 1$ in (11) select -1 . Thus g_0 is the inverse of the common generator $s \mapsto s + 1$. Choosing that inverse generator would give $+1$ here and would simultaneously replace T_0 and A_0 by their inverses.

The transformation laws imply that $\tau - s$, μ , and $\beta + \tau$ descend holomorphically across $t_c = 0$. Recall from (18) that the period matrix is $\Omega = [Z \mid I_2]$, where

$$Z(z) = \begin{pmatrix} 6\mu(z) & \tau(z) \\ \beta(z) & \mu(z) \end{pmatrix}.$$

Recall from (7) that

$$\Lambda_{\text{tor}} = \mathbb{Z}\langle \hat{w}, \hat{\delta} \rangle, \quad \bar{\Lambda} := \Lambda / \Lambda_{\text{tor}} = \mathbb{Z}\langle \hat{\gamma}, \hat{u} \rangle \simeq \mathbb{Z}^2.$$

With respect to the ordered basis $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta})$ of Λ , the period matrix in (18) is

$$\Omega(z) = \left[\begin{array}{cc|cc} 6\mu(z) & \tau(z) & 1 & 0 \\ \beta(z) & \mu(z) & 0 & 1 \end{array} \right] = [Z(z) \mid I_2].$$

Its first two columns are the periods of $\hat{\gamma}, \hat{u}$, whose classes form the displayed quotient basis of $\bar{\Lambda}$; its last two columns are the periods of $\hat{w}, \hat{\delta}$ in Λ_{tor} . Hence Z is exactly the nonconstant 2×2 block of the period matrix, expressed in the bases used to write B_0 above. Its singular part is $s \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = sB_0$; indeed,

$$Z - sB_0 = \begin{pmatrix} 6\mu & \tau - s \\ \beta + s & \mu \end{pmatrix}, \quad \beta + s = (\beta + \tau) - (\tau - s).$$

Every entry on the right descends holomorphically across $t_c = 0$. Consequently, on a small disc $\Delta = \{|t_c| < r\}$,

$$Z = sB_0 + C(t_c), \quad C \in \mathcal{O}(\Delta, M_2(\mathbb{C})). \quad (22)$$

Here is the geometric meaning of this formula. Since the last two columns of $\Omega = [Z \mid I_2]$ represent Λ_{tor} , division by this invariant rank-two sublattice first gives

$$\mathbb{C}^2 / \Omega(\Lambda_{\text{tor}}) \simeq (\mathbb{C}^*)^2, \quad (\zeta_1, \zeta_2) \mapsto (e^{2\pi i \zeta_1}, e^{2\pi i \zeta_2}).$$

The remaining class $\bar{\lambda} \in \bar{\Lambda}$ acts on $(\mathbb{C}^*)^2$ by multiplication with $\exp(2\pi i Z \bar{\lambda})$. Equation (22) turns this multiplier into

$$\exp(2\pi i Z \bar{\lambda}) = t_c^{B_0 \bar{\lambda}} c_{\bar{\lambda}}(t_c), \quad c_{\bar{\lambda}}(t_c) = \exp(2\pi i C(t_c) \bar{\lambda}). \quad (23)$$

If $B_0 \bar{\lambda} = (n_1, n_2)$, the notation $t_c^{B_0 \bar{\lambda}}$ means $(t_c^{n_1}, t_c^{n_2})$; exponentials of vectors are likewise taken componentwise. Thus the singular part of the periods is integral and monomial, while $c_{\bar{\lambda}}$ is a holomorphic nowhere-vanishing correction. This elementary exponential identity is the bridge from the additive period matrix to the toric construction.

The semistable toric model

The monomials in (23) point directly to a toric model. A single cone would provide only one local chart and would not be stable under all translations by $B_0\bar{\Lambda} \simeq \mathbb{Z}^2$. We therefore need a smooth fan invariant under a full rank-two translation lattice. Such a fan is necessarily infinite upstairs; its quotient will nevertheless have finitely many cells and hence a compact central fibre. Concretely, periodically triangulate $\mathbb{R}^2$ by the two standard unimodular triangles in each unit square and cone the resulting cells from the origin in $\mathbb{R}^2 \oplus \mathbb{R}$. The associated locally finite fan $\mathcal{F}$ is smooth, and its toric variety $Y_{\mathcal{F}}$ is a Hausdorff complex threefold. The height character defines $t_c : Y_{\mathcal{F}} \rightarrow \mathbb{C}$; in every maximal toric chart it has the form described next. Let $v_0, v_1, v_2 \in \mathbb{Z}^2 \oplus \mathbb{Z}$ be the primitive generators of a maximal cone. They are the three vertices of one unimodular triangle, each placed at height one. Smoothness means that v_0, v_1, v_2 form a lattice basis. If m_0, m_1, m_2 is the dual basis, then the corresponding affine toric chart is

$$U_{\sigma} \simeq \mathbb{C}^3, \quad z_k := \chi^{m_k} \quad (k = 0, 1, 2),$$

where χ^{m_k} denotes the torus character associated with m_k . The height character takes the value 1 on each v_k , so in the dual basis it is $m_0 + m_1 + m_2$. Its monomial in these coordinates is therefore

$$t_c = z_0 z_1 z_2. \quad (24)$$

Hence the central fibre is a reduced simple-normal-crossing divisor. Each of its components is the toric del Pezzo surface dP_6 , with boundary the hexagon of (-1) -curves. Appendix H gives the exact fan, the determinant-one chart calculation, and the separation argument; the standard toric dictionary is in [8, Sections 1.1–1.4].

Extension of the residual lattice action

Write $Y = Y_{\mathcal{F}}$, and denote by $x = (x_1, x_2)$ the multiplicative coordinates on the dense two-dimensional torus in a fibre over $t_c \neq 0$. Products of such vectors are understood componentwise. Then, for $\bar{\lambda} \in \bar{\Lambda}$, translation of the fan by $B_0\bar{\lambda}$ and multiplication by the holomorphic unit $c_{\bar{\lambda}}(t_c)$ extend the residual lattice action to Y :

$$\Psi_{\bar{\lambda}}(x, t_c) = (c_{\bar{\lambda}}(t_c) t_c^{B_0\bar{\lambda}} x, t_c). \quad (25)$$

The additivity of B_0 and $C(t_c)$ makes this an honest $\bar{\Lambda}$ -action.

Freeness, proper discontinuity, and properness

Introduce the logarithmic modulus coordinate

$$y(x, t_c) = \frac{\log |x|}{\log |t_c|} \in \mathbb{R}^2.$$

Formula (25) becomes the transparent translation law

$$y(\Psi_{\bar{\lambda}}(x, t_c)) = y(x, t_c) + B_{t_c}\bar{\lambda}. \quad (26)$$

Here $B_{t_c} = B_0 - 2\pi \operatorname{Im} C(t_c) / \log |t_c|$ tends to the unimodular matrix B_0 . Uniform invertibility of B_{t_c} turns a fundamental square in $\mathbb{R}^2$ into a uniformly bounded family of parallelograms. Together with local finiteness of the fan, this gives freeness, proper discontinuity, and a compact set meeting every orbit over a closed smaller disc. The complete estimates, including the central fibre, are in Appendix H.

Consequently

$$f_0 : N_0 = Y_\epsilon / \bar{\Lambda} \longrightarrow \Delta_\epsilon := \{|t_c| < \epsilon\}$$

is a proper holomorphic map from a Hausdorff complex threefold. This is the point at which the infinite fan is turned into a compact degeneration; proper discontinuity alone would not suffice.

The special fibre and agreement over the punctured disc

Let $W = f_0^{-1}(0)$. The quotient map is locally biholomorphic, so (24) descends, making W a reduced normal-crossing surface. Unimodularity of B_0 identifies all fan vertices in one orbit. Thus the normalization of W is one copy of dP_6 ; the three pairs of opposite sides of its boundary hexagon are identified, giving three double curves and two triple points. In particular W is compact, connected and irreducible, and $e(W) = 2$. Appendix H derives this description from the cell orbits.

To identify the punctured model with the original family, exponentiate the two fibre coordinates and the additive cusp coordinate:

$$E_{\text{exp}}(z, \zeta) = (e^{2\pi i \zeta_1}, e^{2\pi i \zeta_2}, e^{2\pi i s(z)}) \in (\mathbb{C}^*)^2 \times \mathbb{C}^*.$$

Equation (23) gives the equivariance identity

$$E_{\text{exp}}(z, \zeta + \Omega(z)\lambda) = \Psi_{\bar{\lambda}}(E_{\text{exp}}(z, \zeta)),$$

where $\bar{\lambda}$ denotes the class of λ in $\bar{\Lambda}$. It follows that E_{exp} descends to a biholomorphism

$$G_0 : J|_{0 < |t_c| < \epsilon} \xrightarrow{\sim} N_0 \setminus W. \quad (27)$$

Thus the toric quotient genuinely fills the given family. On the relevant toric affine chart use coordinates

$$(u_0, u_1, u_2) = \left(\frac{t_c}{x_1 x_2}, x_1, x_2 \right).$$

The zero section $\zeta = 0$ is sent by E_{exp} to $(t_c, 1, 1)$ in these coordinates. It therefore extends holomorphically across $t_c = 0$ and meets the smooth locus of W transversely.

6.2 The elliptic points and logarithmic transforms

Cyclic base change and the affine quotient

Fix $j \in \{1, 2\}$ and recall from (4) that g_j has order m_j , where $(m_1, m_2) = (3, 4)$. Choose a small g_j -invariant disc $\Delta_j \subset \mathfrak{h}_z$ centered at its fixed point z_j , and choose the linearizing coordinate s_j from Section 5 so that

$$s_j(g_j z) = e^{-2\pi i / m_j} s_j(z).$$

After multiplying a coarse local coordinate by a holomorphic unit, we may write the quotient map as

$$t_j \circ \pi = s_j^{m_j}.$$

Here t_1 is a local coordinate proportional to t at p_1 , and t_2 is one proportional to $t - 1$ at p_2 . Thus $s_j \mapsto t_j = s_j^{m_j}$ is the cyclic base change that removes the finite orbifold monodromy.

Let $J'_j = (\Delta_j \times \mathbb{C}^2)/\Lambda$ be the pulled-back torus family, where

$$\lambda \cdot (z, \zeta) = (z, \zeta + \Omega(z)\lambda).$$

The cyclic base change has unwrapped the orbifold base coordinate, but we still have to divide by a lift of the deck transformation g_j . The purely linear lift is the obvious first attempt; it has fixed points on the central torus and would create quotient singularities. The remedy is to add a fractional fibre translation and then choose that translation so that the lift still has exact order m_j .

For $b \in \Lambda \otimes \mathbb{R}$, the formula

$$\tilde{g}_j^b(z, \zeta) = (g_j z, R_{g_j}(z)\zeta + \Omega(g_j z)b)$$

defines a biholomorphism of J'_j because of the equivariance relation (21). In the flat real coordinate $x \in (\Lambda \otimes \mathbb{R})/\Lambda$ on the fibre it reads simply

$$(z, x) \mapsto (g_j z, A_j x + b).$$

If $b = v/m_j$ with $A_j v = v$, then the translation parts of the first m_j iterates add to the integral period v , so the resulting torus action has order m_j . If, in addition, the invariant character $\gamma(v)$ is relatively prime to m_j , no nontrivial iterate can acquire a fixed point. These are exactly the two constraints for which the twist vectors were chosen in Section 4.3.

Recall the twist vectors from (8):

$$v_1 = \hat{\gamma} + 2\hat{u} - 4\hat{w}, \quad v_2 = -\hat{\gamma} - 3\hat{u} + 3\hat{w}.$$

They satisfy $A_j v_j = v_j$ and $\gamma(v_1) = 1$, $\gamma(v_2) = -1$. Set

$$\tilde{g}_j = \tilde{g}_j^{v_j/m_j}, \quad \tilde{g}_j(z, \zeta) = \left(g_j z, R_{g_j}(z)\zeta + \Omega(g_j z) \frac{v_j}{m_j} \right), \quad (28)$$

Its m_j th power translates the flat coordinate by

$$\frac{1}{m_j} \sum_{r=0}^{m_j-1} A_j^r v_j = v_j.$$

Translation by $v_j \in \Lambda$ is the identity on the torus. Hence $\tilde{g}_j$ generates a cyclic action of order m_j on J'_j .

Why the affine action is free

The translation by v_j/m_j is essential. Without it, the finite linear monodromy would have fixed points on the central torus and its quotient would be singular. A nonidentity power can have a fixed point only on the central torus. There its translation part is kv_j/m_j , while the integral character γ is invariant under the linear part. But

$$\gamma\left(\frac{kv_1}{3}\right) = \frac{k}{3} \quad (1 \leq k < 3), \quad \gamma\left(\frac{kv_2}{4}\right) = -\frac{k}{4} \quad (1 \leq k < 4).$$

These values are never integral, so the affine fixed-point criterion in Appendix I excludes every fixed point. The quotients

$$N_j = J'_j / \langle \tilde{g}_j \rangle, \quad f_j : N_j \longrightarrow D_j = \Delta_j / \langle g_j \rangle$$

are consequently smooth Hausdorff complex threefolds. Since the pulled-back torus family is proper and the quotient group is finite, each f_j is proper.

The multiple fibres and their surface type

Let $S_j \subset N_j$ be the image of the central torus over $s_j = 0$. The action on that torus is free, so S_j is smooth and admits a finite étale cover by a complex two-torus. Its first rational cohomology is the invariant part of that of the covering torus. Directly from the matrices in (6b),

$$V^{T_1} = \mathbb{Z}\langle \gamma, 2u + w + 3\delta \rangle, \quad V^{T_2} = \mathbb{Z}\langle \gamma, u + w + 2\delta \rangle.$$

Thus $b_1(S_j) = 2$, whereas a complex two-torus has first Betti number 4. The standard classification of free finite quotients of complex two-tori then identifies S_j as a bielliptic surface [9, Chapter VI]. No classification theorem is needed for the later calculations beyond this terminology: the finite torus cover, its affine action, and the invariant lattices have all been displayed explicitly.

The quotient base coordinate is $t_j = s_j^{m_j}$. Pulling the divisor $(t_j = 0)$ back to the cyclic cover gives $m_j(s_j = 0)$. Because the quotient of the total space is locally biholomorphic, this multiplicity descends, and

$$f_1^*(p_1) = 3S_1, \quad f_2^*(p_2) = 4S_2.$$

Thus $3S_1$ and $4S_2$ denote one smooth reduced surface with multiplicity 3 or 4, not three or four irreducible components.

The punctured identification: the logarithmic transform

It remains to compare $N_j \setminus S_j$ with the original family. Write $D_j^* = D_j \setminus \{p_j\}$. On a sector of the punctured s_j -disc, translation in the fibre by

$$\sigma_j(z) = \frac{\log s_j(z)}{2\pi i} \Omega(z) v_j$$

conjugates the affine action (28) to the original linear monodromy action. Although a logarithm has been chosen in this formula, changing its branch adds the integral period $n\Omega(z)v_j$; the induced translation of the torus is therefore unchanged. The sectorial maps hence glue to a single-valued biholomorphism

$$G_j : N_j \setminus S_j \xrightarrow{\sim} J|_{D_j^*}. \quad (29)$$

This operation is the local logarithmic transform: one keeps the torus bundle over the punctured disc but changes how its meridian is completed, (28) records the modification; Appendix I checks the conjugation formula and the gluing map in full.

7 The compact complex threefold

All three local models were built with punctured identifications back to the same torus family J . Consequently the global step introduces no new local formula: it is a gluing problem. What still needs checking is that the glued space is Hausdorff and that the resulting map is proper; without those two points, compatible complex charts alone would not produce a compact complex manifold.

Choose disjoint discs around the three special points. Glue the cusp model N_0 , the logarithmic models N_1, N_2 , and the smooth family J using the punctured identifications above. The result is a complex manifold X with a holomorphic map $f : X \rightarrow \mathbb{P}^1$.

There are two global separation checks.

- Points with different base values are separated by inverse images of disjoint base neighborhoods.
- Points over the same base value lie in one of the Hausdorff local models.

Therefore X is Hausdorff. Properness is local on the base for maps to a Hausdorff space; each of the four local models is proper, and the base is compact. Hence X is compact.

Theorem 7.1 (Analytic output of the construction). *There is a connected compact complex threefold X and a proper surjective holomorphic map $f : X \rightarrow \mathbb{P}^1$ whose general fibre is a complex two-torus, whose cusp fibre is the reduced normal-crossing surface W , and whose two elliptic fibres are $3S_1$ and $4S_2$.*

Up to this point, no topology of S^6 has been used. We have constructed a complex manifold. It remains to recognize its underlying smooth manifold.

8 The fundamental group

We compute the group from the same core-and-caps decomposition used to build X . The only topological input is the Seifert–van Kampen theorem [11, Section 1.2]; the work lies in writing the attaching maps with their correct integral translations.

The base $B^\circ = \mathbb{P}^1 \setminus \{p_0, p_1, p_2\}$ is a pair of pants. Its fundamental group is the free group

$$F_2 = \langle \alpha_1, \alpha_2 \rangle,$$

where α_j is a clockwise meridian around p_j and the clockwise cusp meridian is $\alpha_0 = (\alpha_1 \alpha_2)^{-1}$. The zero section of the smooth torus family lifts these base loops to loops in J and splits the homotopy exact sequence of the fibration; we use the same symbols α_j for these lifted loops. The fibre subgroup is $\pi_1(F) = \Lambda$; for clarity, write t_λ for the fibre loop represented by $\lambda \in \Lambda$.

The resulting group is

$$\pi_1(J) = \Lambda \rtimes_A F_2 = \Lambda \rtimes_A \langle \alpha_1, \alpha_2 \rangle. \quad (30)$$

The subscript records the monodromy homomorphism $A : F_2 \rightarrow \text{GL}(\Lambda)$ with $A(\alpha_j) = A_j$. Explicitly, the underlying set of the semidirect product is $\Lambda \times F_2$, and, writing the abelian group Λ additively, its multiplication is

$$(\lambda, g)(\mu, h) = (\lambda + A(g)\mu, gh).$$

Equivalently, it is generated by the commuting fibre loops t_λ and the two base loops, with conjugation relations

$$\alpha_j t_\lambda \alpha_j^{-1} = t_{A_j \lambda} \quad (\lambda \in \Lambda, j = 1, 2). \quad (31)$$

Geometrically, every loop is a fibre loop followed by a lifted base loop; the failure of these two kinds of loops to commute is exactly the monodromy.

The logarithmic fillings impose

$$\alpha_1^3 = t_{v_1}, \quad \alpha_2^4 = t_{v_2}. \quad (32)$$

Indeed, after m_j turns the base coordinate closes, while the m_j fractional fibre translations add to the integral period v_j . The common sign comes from the common clockwise orientation. The cusp attachment is different: it kills the toric sublattice Λ_{tor} from (7) and the cusp meridian $(\alpha_1 \alpha_2)^{-1}$. The normal closure of Λ_{tor} under A_1, A_2 is $L = \ker \gamma$, so the image of the fibre group becomes a central infinite cyclic group generated by the class c of $\hat{\gamma}$.

For a general cusp regluing with integer ℓ_0 , van Kampen gives

$$\pi_1(X) = \langle c, x, y \mid c \text{ central, } xy = c^{\ell_0}, x^3 = c^{\ell_1}, y^4 = c^{\ell_2} \rangle. \quad (33)$$

Eliminate $y = x^{-1}c^{\ell_0}$. The two remaining relations in the generators (x, c) have matrix

$$\begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}.$$

Its first Smith divisor is 1, since the entries include 3 and 4, and its determinant has absolute value $|p|$, where p is (5). Hence

$$\pi_1(X) \simeq \mathbb{Z}/|p|\mathbb{Z}. \quad (34)$$

For $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$, we obtain $p = -1$ and therefore

$$\pi_1(X) = 1.$$

This computation reveals the purpose of the translations in (8): their linear parts control the analytic multiple fibres, while their γ -components kill the final coinvariant circle.

9 Nearby cycles at the cusp

Why is a separate cusp calculation needed after the analytic filling is complete? The Leray spectral sequence requires the actual integral stalk of $R^q f_* \mathbb{Z}$ at p_0 , together with its inclusion in the cohomology of a nearby torus. Monodromy identifies the expected invariant subgroup, but by itself does not rule out a finite-index discrepancy. The global homology computation must therefore be integral. Rational ranks are not enough: an unnoticed index two in a specialization lattice would change a Leray differential from multiplication by p to multiplication by $2p$ or $p/2$. The proof therefore computes both ranks and saturation.

9.1 Local Milnor fibres

At a point where r branches of W meet, the local equation is

$$z_0 \cdots z_{r-1} = t_c.$$

The Milnor fibre retracts onto $(S^1)^{r-1}$. Thus over the smooth locus, the double curves, and the two triple points, the local Milnor fibres are respectively

$$\text{pt}, \quad S^1, \quad T^2.$$

Let D_1, D_2, D_3 be the double curves and P, Q the triple points. The nearby cohomology sheaves are

$$\mathcal{H}^0 \psi = \mathbb{Z}_W, \quad \mathcal{H}^1 \psi = \mathcal{V}_{\text{van}}, \quad \mathcal{H}^2 \psi = \mathbb{Z}_P \oplus \mathbb{Z}_Q.$$

The vanishing-cycle local system has no hidden sign monodromy: in a chart $xy = t_c$, toric translations multiply x and y by reciprocal units and preserve the oriented phase difference $\arg x - \arg y$.

At either triple point, the three restrictions of a class in $H^1(T^2; \mathbb{Z})$ satisfy one relation. With compatible orientations it is $n_1 - n_2 + n_3 = 0$ at both P and Q . Hence

$$0 \rightarrow \mathcal{V}_{\text{van}} \rightarrow \bigoplus_{k=1}^3 \mathbb{Z}_{D_k} \xrightarrow{\delta} \mathbb{Z}_P \oplus \mathbb{Z}_Q \rightarrow 0, \quad \delta(n_1, n_2, n_3) = (s, s), \quad s = n_1 - n_2 + n_3. \quad (35)$$

The normalization $\nu : dP_6 \rightarrow W$ gives a parallel resolution

$$0 \rightarrow \mathbb{Z}_W \rightarrow \nu_* \mathbb{Z}_{dP_6} \rightarrow \bigoplus_{k=1}^3 \mathbb{Z}_{D_k} \xrightarrow{\delta} \mathbb{Z}_P \oplus \mathbb{Z}_Q \rightarrow 0. \quad (36)$$

Using $H^\bullet(dP_6; \mathbb{Z}) = (\mathbb{Z}, 0, \mathbb{Z}^4, 0, \mathbb{Z})$ and $H^\bullet(\mathbb{P}^1; \mathbb{Z}) = (\mathbb{Z}, 0, \mathbb{Z})$ gives

$$H^\bullet(W; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}), \quad (37)$$

$$H^\bullet(W; \mathcal{V}_{\text{van}}) = (\mathbb{Z}^2, \mathbb{Z}, \mathbb{Z}^3, 0). \quad (38)$$

Here a tuple lists cohomology groups in increasing degree. The nearby-cycle spectral sequence

$$E_2^{a,b} = H^a(W, \mathcal{H}^b \psi) \implies H^{a+b}(F_t; \mathbb{Z}) \quad (39)$$

therefore begins with the rank table

$b = 2$	2	0	0	0	0
$b = 1$	2	1	3	0	0
$b = 0$	1	2	4	2	1
	$a = 0$	$a = 1$	$a = 2$	$a = 3$	$a = 4$

It converges to the cohomology of the smooth fibre T^4 , whose Betti numbers are $(1, 4, 6, 4, 1)$.

9.2 The saturation hinge

Rank comparison alone leaves a possible ambiguity in total degree three. Choose a compact real two-torus K in the dense torus of the smooth locus of W . Its nearby translate $K_t \subset F_t$ has fundamental lattice

$$\pi_1(K_t) = \Lambda_{\text{tor}} = \mathbb{Z} \langle \widehat{w}, \widehat{\delta} \rangle,$$

a primitive direct summand of Λ . Thom isomorphism and a product tubular neighborhood give a commutative square

$$\begin{array}{ccc} H^3(W, W \setminus K) & \longrightarrow & H^3(W) \\ \downarrow \wr & & \downarrow \text{sp}^3 \\ H^3(F_t, F_t \setminus K_t) & \longrightarrow & H^3(F_t). \end{array}$$

The bottom horizontal map is the Gysin map

$$H^1(T_f) \longrightarrow H^3(T_f \times T_B), \quad a \longmapsto a \times [T_B]^\vee,$$

so its image is a saturated rank-two direct summand. A calculation with the six boundary curves of dP_6 shows that the supported classes generate all of $H^3(W) \simeq \mathbb{Z}^2$. Consequently specialization in degree three is injective with saturated image. The remaining degrees are then controlled by the free filtration quotients of the nearby-cycle spectral sequence.

The result is the integral invariant-cycle statement

$$H^q(W; \mathbb{Z}) \xrightarrow{\sim} H^q(F; \mathbb{Z})^{T_0}, \quad 0 \leq q \leq 4. \quad (40)$$

This is the most delicate local topological input at the cusp. General nearby-cycle theory explains why specialization lands in monodromy invariants [10, Chapter 4]; the argument above is needed to prove that, in this example, the image is the *entire integral* invariant lattice rather than a finite-index sublattice.

10 The global integral Leray calculation

The local analysis has now converted each singular fibre into explicit integral lattice data. Leray is the assembly mechanism. The guiding idea is to identify primitive generators so that every possible transgression is measured by the same filling integer p from (5). Once p is a unit, the relevant source and target copies of $\mathbb{Z}$ cancel in pairs; the remaining work is to justify that no hidden torsion or middle-column class has been overlooked.

Let $L^q = R^q f_* \mathbb{Z}|_{B^\circ}$, with fibre $\bigwedge^q V$. Equation (40) describes its extension at p_0 . At the elliptic points, the reduced fibres are free cyclic quotients of a smooth torus. Transfer, the group-extension spectral sequence, and the integral intersection pairings give the specialization indices

	$q = 1$	$q = 2$	$q = 3$	$q = 4$
p_1	3	1	1	3
p_2	4	2	2	4

(41)

The index two in degree two at p_2 is especially important. It is not visible from rational cohomology.

We now determine the three columns of the spectral sequence (2). The groups H^0 are global monodromy-invariant classes that extend across every special fibre. The groups H^2 are the dual coinvariant lattices. The middle groups H^1 measure the obstruction to patching local invariant sections on the punctured sphere. Once these three columns are known, only the transgressions d_2 remain.

Put

$$q_0 = u \wedge w + 6\gamma \wedge \delta, \quad \text{vol} = \gamma \wedge u \wedge w \wedge \delta.$$

Integral row reduction gives the global invariant and coinvariant lattices

q	1	2	3	4
$(\bigwedge^q V)^G$	$\mathbb{Z}\gamma$	$\mathbb{Z}q_0$	$\mathbb{Z}(\gamma \wedge u \wedge w)$	$\mathbb{Z}\text{vol}$
$(\bigwedge^q V)_G$	$\mathbb{Z}[\delta]$	$\mathbb{Z}[\gamma \wedge \delta]$	$\mathbb{Z}[u \wedge w \wedge \delta]$	$\mathbb{Z}$

(42)

Intersecting the invariant lines with the local specialization lattices in (41) yields

$$H^0(B, R^1 f_* \mathbb{Z}) = 12\mathbb{Z}\gamma, \quad H^0(B, R^2 f_* \mathbb{Z}) = \mathbb{Z}(2q_0), \quad (43)$$

$$H^0(B, R^3 f_* \mathbb{Z}) = \mathbb{Z}(2\gamma \wedge u \wedge w), \quad H^0(B, R^4 f_* \mathbb{Z}) = 12\mathbb{Z}\text{vol}. \quad (44)$$

Poincaré duality with local coefficients on B° gives

$$H^2(B, R^q f_* \mathbb{Z}) \simeq (\bigwedge^q V)_G.$$

A direct parabolic-cocycle calculation also gives

$$H^1(B, R^1 f_* \mathbb{Z}) = H^1(B, R^2 f_* \mathbb{Z}) = 0. \quad (45)$$

Because the base has complex dimension one, the only possible differentials are

$$d_2 : E_2^{0,q} \longrightarrow E_2^{2,q-1}.$$

The first is computed on an actual core-and-caps cover of the base. A degree-one fibre class 12γ extends over the pair-of-pants core. Over the three caps, its character values on the boundary meridians are

$$12\ell_0, \quad 4\ell_1, \quad 3\ell_2.$$

The Mayer–Vietoris boundary, with the common orientation fixed earlier, therefore gives

$$d_2(12\gamma) = p\eta, \quad p = 12\ell_0 - 4\ell_1 - 3\ell_2, \quad (46)$$

where η generates $H^2(B; \mathbb{Z})$.

Write the next two differentials as

$$d_2(2q_0) = p'\eta[\delta], \quad d_2(2\gamma \wedge u \wedge w) = p''\eta[\gamma \wedge \delta].$$

The Leray spectral sequence is multiplicative and d_2 is a derivation. Using

$$(12\gamma)(2q_0) = 12(2\gamma \wedge u \wedge w), \quad [q_0] = 12[\gamma \wedge \delta],$$

gives $p'' = 2p - p'$. Applying the derivation rule to the vanishing product

$$(2q_0)(2\gamma \wedge u \wedge w) = 0$$

and using the two primitive target pairings gives $p' = p''$. Hence

$$d_2(2q_0) = p\eta[\delta], \quad d_2(2\gamma \wedge u \wedge w) = p\eta[\gamma \wedge \delta]. \quad (47)$$

For the chosen twists, $p = -1$. All three maps (46)–(47) are therefore isomorphisms of infinite cyclic groups. Together with (45), the Leray filtration gives the claimed vanishing as follows. In total degree one, $E_\infty^{0,1}$ is the kernel of the first isomorphism. In total degree two, $E_\infty^{2,0}$ is its cokernel and $E_\infty^{0,2}$ is the kernel of the second. In total degree three, the possible pieces are the cokernel of the second map, the kernel of the third, and the already vanishing middle column. Every piece is zero, so

$$H^1(X; \mathbb{Z}) = H^2(X; \mathbb{Z}) = H^3(X; \mathbb{Z}) = 0. \quad (48)$$

Since $\pi_1(X) = 1$, we have $H_1(X) = 0$. The universal coefficient theorem first shows from $H^2 = 0$ that $H_2(X)$ is torsion; it is finite because X has the homotopy type of a finite CW complex. The theorem then shows from $H^3 = 0$ that $\text{Ext}(H_2(X), \mathbb{Z}) = 0$. A finite abelian group is isomorphic to its own Ext group, so $H_2(X) = 0$. Integral Poincaré duality on the closed oriented real six-manifold X now yields

$$H_k(X; \mathbb{Z}) = \begin{cases} \mathbb{Z}, & k = 0, 6, \\ 0, & 1 \leq k \leq 5. \end{cases} \quad (49)$$

11 Recognition of the standard sphere

We have proved that X is simply connected and has the integral homology of S^6 . Suppose that some $\pi_k(X)$ with $2 \leq k \leq 5$ were nonzero, and choose the least such k . Then X would be $(k-1)$ -connected, so the Hurewicz theorem would identify $\pi_k(X)$ with $H_k(X) = 0$, a contradiction. Applying Hurewicz once more in degree six gives

$$\pi_k(X) = 0 \quad (1 \leq k \leq 5), \quad \pi_6(X) \simeq H_6(X) \simeq \mathbb{Z}.$$

A map $S^6 \rightarrow X$ representing a generator of $\pi_6(X)$ is a homology isomorphism. The homological Whitehead theorem makes it a homotopy equivalence. These forms of Hurewicz and Whitehead are reviewed in [11, Chapter 4]. Thus X is a smooth homotopy six-sphere.

Smooth homotopy n -spheres form the Kervaire–Milnor group Θ_n for $n \geq 5$. The classical computation gives

$$\Theta_6 = 0.$$

Therefore every smooth homotopy six-sphere is diffeomorphic to the standard S^6 ; this is the six-dimensional entry in the computation of Kervaire–Milnor [13], built on the high-dimensional Poincaré theorem of Smale [12]. Consequently there exists a diffeomorphism

$$\Phi : S^6 \longrightarrow X.$$

If (U_α, z_α) is the holomorphic atlas of X , then

$$(\Phi^{-1}(U_\alpha), z_\alpha \circ \Phi)$$

is a complex atlas on S^6 . Its transition functions are the transition functions of X , hence holomorphic.

Theorem 11.1 (Candidate headline theorem). *Subject to the validity of the analytic and integral-topological arguments above, the standard smooth six-sphere admits an integrable complex structure.*

The wording “subject to the validity” is not a mathematical hypothesis inside the proof. It records the external status of an exceptional, unrefereed claim.

12 What the construction does, and what it does not do

12.1 It does not integrate the octonionic structure

The familiar octonionic almost-complex structure is not being repaired. The candidate complex structure is obtained from a torus fibration and is transported to S^6 only after the smooth recognition theorem. There is no reason for it to be orthogonal to the round metric, nearly Kähler, or invariant under the standard G_2 action.

12.2 It is necessarily non-Kähler

Since $H^2(S^6; \mathbb{R}) = 0$, the transported complex manifold cannot carry a Kähler form. The family itself exhibits non-Kähler behavior: it has multiple fibres, a non-normal normal-crossing fibre, and nontrivial monodromy. Familiar Kähler identities should therefore not be imported into its study. General restrictions on a hypothetical complex structure on S^6 are developed in [15]; results on the algebraic dimension of compact complex threefolds with vanishing second Betti number appear in [14]. Those works provide context, but neither is used to construct the manifold X or to compute its topology here.

12.3 The diffeomorphism is existential

The proof recognizes X by homotopy-sphere theory. It does not give an explicit coordinate formula for $\Phi : S^6 \rightarrow X$, and therefore does not produce a simple 6×6 matrix formula for the resulting tensor on the round sphere. This is not a logical gap: complex structures transport through any diffeomorphism. It is, however, a limitation for explicit geometry.

12.4 The three special fibres do distinct jobs

- The cusp fibre supplies vanishing cycles and carries the entire Euler characteristic.
- The order-3 and order-4 logarithmic fibres supply the arithmetic coefficients 4 and 3 in the twist integer p .
- Their common interaction with the global invariant class produces the coefficient $12 = \text{lcm}(3, 4)$.

Thus the formula $p = 12\ell_0 - 4\ell_1 - 3\ell_2$ is the meeting point of the geometry and the topology, not an accidental computation.

13 Audit map: where a verifier should look

For an extraordinary claim, a reader should know which steps are routine and which are load bearing.

Stage	What must be checked	Failure mode
Modular lift	Holomorphic solution of $j \circ \tau = 1728t \circ \pi$; simultaneous normalization of the two elliptic transformations	Independent local normalizations could give the wrong cusp element
Affine periods	Exact torsor sheaves $\mathcal{O}(-1)$ and $\mathcal{O}$; multiplier of $E_6^{1/2}$	A sign error changes equivariance of the lattice
Period lattice	Nonvanishing real determinant $\text{Im } \tau D$	The quotient would fail to be a compact torus
Cusp quotient	Freeness, proper discontinuity, Hausdorffness, and compact fundamental domains as $t_c \rightarrow 0$	An infinite fan can produce a non-Hausdorff or nonproper quotient
Log transforms	Freeness of every nontrivial power and common orientation of the two elliptic generators	Orbifold singularities or wrong signs in p
Nearby cycles	Orientation of the vanishing local systems and integral saturation of specialization	Hidden 2-torsion or an incorrect Leray coefficient
Elliptic lattices	The index-two degree-two lattice at p_2	Replacing p by $2p$ or $p/2$
Leray products	Primitive generators and the signs in the derivation rule	Failure of the three differentials to have the same coefficient
Recognition	Integral, not merely rational, homology; use of $\Theta_6 = 0$	A homology manifold need not be the standard smooth sphere

The attached 28-page reconstruction explicitly identified properness of the infinite-fan quotient and integral saturation at the cusp as the two highest-risk points.

14 A compact study guide

The construction becomes easier to remember if every object is assigned one job.

Object	Job
T_1, T_2, T_0	finite monodromies at 3, 4 and unipotent monodromy at the cusp
τ	modular lift matching the two finite branch orders and the cusp
μ, β	affine corrections forced by descent of the normalized period lattice
$D < 0$	certifies that the four columns of Ω form a real lattice
B_0	converts the unipotent monodromy into translations of the toric fan
v_1, v_2	make the elliptic actions free and fix the twist numbers
γ	detects the sole global fibre coinvariant
W	supplies the cusp degeneration and the nearby-cycle complex
p	simultaneously controls π_1 and the three Leray transgressions
$\Theta_6 = 0$	converts a smooth homotopy sphere into the standard sphere

For a first seminar talk, stop after the fundamental-group computation and state the integral Leray calculation as a theorem. For a reading course, work through Sections 9 and 10 separately: the first is local degeneration theory, and the second is global sheaf cohomology on the base. For a verification project, reproduce the Smith normal forms of (42) and the index-two lattice in (41) before attempting the spectral sequence.

A The geometry of the orbifold $(0; 3, 4, \infty)$

This appendix expands the geometric picture summarized in Section 3. The signature

$$(0; 3, 4, \infty)$$

contains four pieces of information. The initial 0 is the genus of the coarse compact Riemann surface, so that surface is $B \simeq \mathbb{P}^1$. The entries 3 and 4 specify cone points at p_1, p_2 , while ∞ specifies a cusp at p_0 .

For a cone point of order m , let a cyclic group act on a disc by

$$s \mapsto e^{-2\pi i/m} s.$$

The invariant coordinate is $u = s^m$. Hence the coarse quotient is again a smooth disc, but the orbifold remembers the stabilizer and the quotient cone angle $2\pi/m$. This gives angles $2\pi/3$ at p_1 and $\pi/2$ at p_2 . It also explains why the quotient map used later has local orders 3 and 4.

The cusp is not a finite quotient. Conjugate its stabilizer so that a generator acts on the upper half-plane by $w \mapsto w - 1$. Recall from (3) that the high horodiscs centred at this ideal point are $\mathcal{H}_Y = \{\operatorname{Im} w > Y\}$. Then

$$q = e^{2\pi i w}$$

is invariant and satisfies $|q| = e^{-2\pi \operatorname{Im} w}$. Consequently

$$\mathcal{H}_Y / \langle w \mapsto w - 1 \rangle \simeq \{0 < |q| < e^{-2\pi Y}\};$$

adjoining $q = 0$ fills the puncture. Thus the symbol ∞ means parabolic monodromy of infinite order.

The orbifold is hyperbolic because

$$\chi_{\text{orb}} = 2 - \left(1 - \frac{1}{3}\right) - \left(1 - \frac{1}{4}\right) - 1 = -\frac{5}{12} < 0.$$

Equivalently, the hyperbolic upper half-plane contains a geodesic triangle whose angles are

$$\frac{\pi}{3}, \quad \frac{\pi}{4}, \quad 0.$$

The zero-angle vertex is ideal. Reflections in its sides tile the upper half-plane. The orientation-preserving subgroup has as a fundamental region two adjacent triangles. Composing the side reflections at the two finite vertices gives clockwise rotations g_1, g_2 of orders 3, 4. With the compatible orientation at the ideal vertex,

$$g_0 = (g_1 g_2)^{-1}$$

is parabolic. This proves the presentation

$$\Gamma_{\text{orb}} = \langle g_1, g_2 \mid g_1^3 = g_2^4 = 1 \rangle$$

and identifies the quotient with $B \setminus \{p_0\}$. The orientation choice is not cosmetic: it fixes the sign of the cusp relation used later in van Kampen and in the Mayer–Vietoris computation of the Leray differential. See [5] for the general triangle-group theory.

B The modular invariant and its local branches

Put $q = e^{2\pi i\tau}$ and $\sigma_r(n) = \sum_{d|n} d^r$. The normalized Eisenstein series are

$$E_4(\tau) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n, \quad E_6(\tau) = 1 - 504 \sum_{n \geq 1} \sigma_5(n) q^n.$$

For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ they satisfy

$$E_k(M\tau) = (c\tau + d)^k E_k(\tau), \quad k = 4, 6.$$

The discriminant

$$\Delta(\tau) = \frac{E_4(\tau)^3 - E_6(\tau)^2}{1728} = q \prod_{n \geq 1} (1 - q^n)^{24}$$

has weight 12 and no zero on $\mathfrak{h}_\tau$. Therefore

$$j = \frac{E_4^3}{\Delta}$$

is modular invariant. Its Fourier expansion is $j = q^{-1} + 744 + 196884q + O(q^2)$, so it has a simple pole at the cusp. It induces

$$\mathrm{PSL}_2(\mathbb{Z}) \backslash \mathfrak{h}_\tau \simeq \mathbb{C}_j, \quad \mathrm{PSL}_2(\mathbb{Z}) \backslash \mathfrak{h}_\tau^* \simeq \mathbb{P}_j^1,$$

where $\mathfrak{h}_\tau^* = \mathfrak{h}_\tau \cup \mathbb{P}^1(\mathbb{Q})$ and all rational boundary points form one cusp orbit.

There are two exceptional orbits in $\mathfrak{h}_\tau$. For $\rho = e^{2\pi i/3}$, the stabilizer of ρ in $\mathrm{PSL}_2(\mathbb{Z})$ has order 3; the stabilizer of i has order 2. The associated lattices are hexagonal and square. Multiplication by ρ and by i gives their extra automorphisms; quotienting the common central automorphism -1 reduces the corresponding orders from 6, 4 to 3, 2.

The valence formula shows that E_4 has a simple zero at the orbit of ρ and E_6 a simple zero at the orbit of i . Since

$$j - 1728 = \frac{E_6^2}{\Delta},$$

one obtains, for nonzero constants a_3, a_2 ,

$$j(\tau) = a_3(\tau - \rho)^3 + O((\tau - \rho)^4), \quad j(\tau) - 1728 = a_2(\tau - i)^2 + O((\tau - i)^3).$$

Thus the invariant map from $\mathfrak{h}_\tau$ has ramification indices 3 over 0 and 2 over 1728. By contrast, after passing to the coarse quotient, j itself is the global coordinate. These facts are standard; see [6, Chapters 1–3].

C The real determinant of the period matrix

We spell out the sign and block calculation behind (19). Give $\mathbb{C}^2$ its complex orientation, with ordered real coordinates

$$(\mathrm{Re} z_1, \mathrm{Im} z_1, \mathrm{Re} z_2, \mathrm{Im} z_2).$$

In the ordered basis $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta})$ of Λ , the real matrix of $\Omega = [Z \mid I_2]$ is

$$M_{\Omega, \mathbb{R}} = \begin{pmatrix} \mathrm{Re}(6\mu) & \mathrm{Re} \tau & 1 & 0 \\ \mathrm{Im}(6\mu) & \mathrm{Im} \tau & 0 & 0 \\ \mathrm{Re} \beta & \mathrm{Re} \mu & 0 & 1 \\ \mathrm{Im} \beta & \mathrm{Im} \mu & 0 & 0 \end{pmatrix}.$$

Temporarily reorder the target coordinates as $(\operatorname{Re} z_1, \operatorname{Re} z_2, \operatorname{Im} z_1, \operatorname{Im} z_2)$. This is one row transposition, and the matrix becomes

$$\begin{pmatrix} \operatorname{Re} Z & I_2 \\ \operatorname{Im} Z & 0 \end{pmatrix}.$$

Interchanging the two blocks of two columns has sign $(-1)^{2 \cdot 2} = 1$ and gives the block upper-triangular matrix

$$\begin{pmatrix} I_2 & \operatorname{Re} Z \\ 0 & \operatorname{Im} Z \end{pmatrix}.$$

Undoing the single row transposition therefore yields

$$\det_{\mathbb{R}} \Omega = -\det(\operatorname{Im} Z).$$

Since

$$\operatorname{Im} Z = \begin{pmatrix} 6 \operatorname{Im} \mu & \operatorname{Im} \tau \\ \operatorname{Im} \beta & \operatorname{Im} \mu \end{pmatrix},$$

we conclude that

$$\begin{aligned} \det_{\mathbb{R}} \Omega &= -\left(6(\operatorname{Im} \mu)^2 - \operatorname{Im} \tau \operatorname{Im} \beta\right) \\ &= \operatorname{Im} \tau \left(\operatorname{Im} \beta - \frac{6(\operatorname{Im} \mu)^2}{\operatorname{Im} \tau}\right) \\ &= \operatorname{Im} \tau D. \end{aligned}$$

Because $\operatorname{Im} \tau > 0$ and $D < 0$, this determinant is nonzero. Hence the four columns of Ω span a full real lattice in $\mathbb{C}^2$.

D Background results used in the final recognition

For reference, here are the precise classical statements used after the analytic construction.

Theorem D.1 (Hurewicz, inductive form). *If a path-connected CW complex is $(n-1)$ -connected, then its reduced homology vanishes below degree n and the Hurewicz map $\pi_n \rightarrow H_n(-; \mathbb{Z})$ is an isomorphism.*

Theorem D.2 (Homological Whitehead). *A homology isomorphism between simply connected CW complexes is a homotopy equivalence.*

Both statements are classical; see [11, Chapter 4].

Theorem D.3 (Universal coefficient and Poincaré duality). *For a CW complex there is an exact sequence*

$$0 \rightarrow \operatorname{Ext}(H_{k-1}, \mathbb{Z}) \rightarrow H^k(-; \mathbb{Z}) \rightarrow \operatorname{Hom}(H_k, \mathbb{Z}) \rightarrow 0.$$

For a closed oriented six-manifold, cap product with the fundamental class identifies H^k with H_{6-k} integrally.

The universal coefficient theorem and Poincaré duality in this form are reviewed in [11, Chapter 3].

Theorem D.4 (Smooth six-spheres). *The group of oriented diffeomorphism classes of smooth homotopy six-spheres is trivial: $\Theta_6 = 0$.*

The high-dimensional recognition input is due to Smale [12], and the group of smooth homotopy spheres is computed by Kervaire–Milnor [13].

E Exact integral algebra behind the monodromy

This appendix supplies the integer calculations used in Sections 4 and 10. All lattices below are written in displayed ordered bases. No passage to $\mathbb{Q}$ is used when a saturation or an index is at issue.

E.1 The homological matrices

The inverse-transpose matrices acting on $\Lambda = \mathbb{Z}\langle\widehat{\gamma}, \widehat{u}, \widehat{w}, \widehat{\delta}\rangle$ are

$$A_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 6 & 0 & 1 & 0 \\ -6 & -1 & -1 & 0 \\ -2 & 1 & 0 & 1 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ -6 & 1 & 0 & 0 \\ 3 & 0 & 1 & 1 \end{pmatrix}.$$

Multiplication gives $A_1^3 = A_2^4 = I$. On the last three basis vectors,

$$\begin{aligned} A_1\widehat{u} &= -\widehat{w} + \widehat{\delta}, & A_1\widehat{w} &= \widehat{u} - \widehat{w}, & A_1\widehat{\delta} &= \widehat{\delta}, \\ A_2\widehat{u} &= \widehat{w}, & A_2\widehat{w} &= -\widehat{u} + \widehat{\delta}, & A_2\widehat{\delta} &= \widehat{\delta}. \end{aligned}$$

Since $A_0 = (A_1 A_2)^{-1}$, direct multiplication also gives

$$A_0 - I = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{pmatrix}.$$

It follows at once that

$$\ker(A_0 - I) = \text{im}(A_0 - I) = \mathbb{Z}\langle\widehat{w}, \widehat{\delta}\rangle.$$

Modulo this common lattice, $A_0 - I$ sends $\widehat{\gamma}$ to $-\widehat{\delta}$ and $\widehat{u}$ to $\widehat{w}$. Thus the induced matrix B_0 has determinant one. This verifies both the equality in (7) and the unimodularity assertion following it.

E.2 Global fibre coinvariants

The images of $A_1 - I$ and $A_2 - I$ lie in $L = \ker \gamma$. Conversely, the three vectors

$$(A_1 - I)\widehat{u} = -\widehat{u} - \widehat{w} + \widehat{\delta}, \quad (A_1 - I)\widehat{w} = \widehat{u} - 2\widehat{w}, \quad (A_2 - I)\widehat{u} = -\widehat{u} + \widehat{w}$$

together with $(A_2 - I)\widehat{w} = -\widehat{u} - \widehat{w} + \widehat{\delta}$ generate a rank-three sublattice of L . A 3×3 minor of their coordinate matrix in $(\widehat{u}, \widehat{w}, \widehat{\delta})$ is ± 1 . Hence the generated lattice is saturated and equals L . Therefore

$$\Lambda_G := \Lambda / ((A_1 - I)\Lambda + (A_2 - I)\Lambda) \simeq \Lambda / L \simeq \mathbb{Z},$$

with primitive coordinate γ .

For the twists

$$v_1 = \widehat{\gamma} + 2\widehat{u} - 4\widehat{w}, \quad v_2 = -\widehat{\gamma} - 3\widehat{u} + 3\widehat{w},$$

matrix multiplication gives $A_1 v_1 = v_1$, $A_2 v_2 = v_2$, and evaluation gives $\gamma(v_1) = 1$, $\gamma(v_2) = -1$. These four equalities are the entire arithmetic input to the finite affine actions.

E.3 Exterior powers

On the dual lattice V , the images in degree one are

$$(T_1 - I)V = \mathbb{Z}\langle 2\gamma + u + w, 2\gamma - u \rangle, \quad (T_2 - I)V = \mathbb{Z}\langle 3\gamma + u, u + w \rangle.$$

Their sum is the saturated lattice $\mathbb{Z}\langle \gamma, u, w \rangle$; the common invariant lattice is $\mathbb{Z}\gamma$. In degrees two and three, use the ordered bases

$$(\gamma u, \gamma w, \gamma \delta, uw, u\delta, w\delta), \quad (\gamma uw, \gamma u\delta, \gamma w\delta, uw\delta),$$

where juxtaposition means wedge product. Integer row reduction of the stacked matrices for $\bigwedge^q T_1 - I$ and $\bigwedge^q T_2 - I$ has Smith forms

$$\text{diag}(1, 1, 1, 1, 1, 0) \quad (q = 2), \quad \text{diag}(1, 1, 1, 0) \quad (q = 3).$$

Solving the corresponding kernel equations gives exactly (42). Two relations needed later are

$$[u \wedge w] = 6[\gamma \wedge \delta], \quad [q_0] = 12[\gamma \wedge \delta]$$

in the degree-two coinvariants. Since the Smith forms have only unit nonzero entries, all displayed invariant and coinvariant generators are primitive. This is the integral information that rational row reduction would miss.

F Construction of the three period functions

We now prove the existence assertions of Section 5. The argument separates the modular lift τ from the two affine functions μ and β .

F.1 The modular lift and its normalization

Remove the inverse images of p_1, p_2 from $\mathfrak{h}_z$. The restriction

$$j : \mathfrak{h}_\tau \setminus j^{-1}\{0, 1728\} \longrightarrow \mathbb{C} \setminus \{0, 1728\}$$

is an unramified covering. A local inverse of j therefore continues along paths. A small meridian over p_1 maps under $1728t \circ \pi$ to the third power of a meridian about 0, exactly cancelling the ramification index three of j . A small meridian over p_2 maps to the fourth power of a meridian about 1728; this is divisible by the ramification index two of j . Thus all continuations around the deleted points close. These local checks also control arbitrary loops: because $\mathfrak{h}_z$ is simply connected, a loop in the complement of the discrete inverse images of p_1, p_2 bounds a disc; after making that disc transverse to the discrete set, its boundary is a product of conjugates of finitely many small meridians. Hence the pullback of the unramified modular covering has trivial monodromy, and the inverse is single valued.

The extension across the deleted points is also explicit. If s_1 is a linearizing coordinate at a lift of p_1 , then

$$t \circ \pi = s_1^3 u_1(s_1), \quad u_1(0) \neq 0,$$

whereas the T -translate of (10) gives $j(\tau) = a_3'(\tau - (\rho + 1))^3 + \dots$. A holomorphic cube root of the unit u_1 gives $\tau - (\rho + 1)$ as a holomorphic function vanishing to order one. Similarly,

$$(t - 1) \circ \pi = s_2^4 u_2(s_2), \quad j(\tau) - 1728 = a_2(\tau - i)^2 + \dots,$$

so a square root gives an extension for which $\tau - i$ vanishes to order two. Hence there is a holomorphic map $\tau : \mathfrak{h}_z \rightarrow \mathfrak{h}_\tau$ satisfying $j(\tau) = 1728t \circ \pi$.

For $g \in \Gamma_{\text{orb}}$, the lifts $\tau \circ g$ and τ differ by one element of $\text{PSL}_2(\mathbb{Z})$. This defines a single monodromy representation. The images of g_1, g_2 have orders three and two in the modular group. Their product is the positively oriented primitive parabolic. To normalize both simultaneously, choose lifts $A, B \in \text{SL}_2(\mathbb{Z})$ with $AB = T$ and $\text{tr } A = 1$. Write $A = \begin{pmatrix} r & s \\ c & d \end{pmatrix}$. Since $B = A^{-1}T$ has trace zero, $c = 1$; since $r + d = 1$ and $\det A = 1$, $s = r(1 - r) - 1$. Conjugating by T^{1-r} gives

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}, \quad B \equiv \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

This proves all three transformation laws in (11) with a single global choice.

F.2 The affine torsor for μ

The difference φ of two local solutions for μ obeys

$$\varphi \circ g_1 = -\varphi/\tau, \quad \varphi \circ g_2 = \varphi/\tau.$$

Because the zeros of $E_6 \circ \tau$ have even order, this function has a holomorphic square root on the simply connected domain $\mathfrak{h}_z$. Put

$$F = \frac{E_4(\tau)^2 E_6(\tau)^{1/2}}{\Delta(\tau)}.$$

The weight is $8 + 3 - 12 = -1$. The square root might a priori introduce a sign in a transformation law. Iteration around the order-three point forces the first sign to satisfy $\epsilon_1^3 = 1$, hence $\epsilon_1 = 1$. At the order-four point, a coordinate with $s_2 \circ g_2 = -is_2$ gives $E_6(\tau)^{1/2} = cs_2 + O(s_2^2)$; consequently $F(g_2 z)/F(z) \rightarrow -i = 1/\tau(z_2)$. Thus the second sign is also fixed.

The orders of F at p_1, p_2, p_0 are $2, 1, -1$. A homogeneous solution φ has the same forced orders at the first two points, while it is bounded at the cusp. Therefore φ/F descends to a holomorphic function on B vanishing at p_0 . Conversely, multiplication by F turns every section of $\mathcal{O}_B(-p_0)$ into a homogeneous solution. The sheaf of differences is exactly $\mathcal{O}_B(-1)$.

Local affine solutions exist: use $(2 - \tau)/3$ at p_1 , $(1 - \tau)/2$ at p_2 , and 0 at the cusp. Their differences on overlaps form a Čech cocycle in $\mathcal{O}(-1)$. Since $H^1(B, \mathcal{O}(-1)) = 0$, the cocycle is a coboundary and the local solutions glue. Since $H^0(B, \mathcal{O}(-1)) = 0$, the resulting μ is unique.

F.3 The affine torsor for β

For a finite cyclic stabilizer, sum the affine increments in (16) around the orbit. Substitution of the transformation law for μ makes the sum zero, so averaging supplies a local fixed solution. At the cusp, composition gives $\tau \circ g_0 = \tau - 1$ and $\beta \circ g_0 = \beta + 1$; hence $\beta = -\tau$ is a local solution. Differences of two solutions are invariant, so the difference sheaf is $\mathcal{O}_B$. The equality $H^1(B, \mathcal{O}_B) = 0$ gives a global solution, unique up to a constant.

Direct substitution shows that D in (17) is invariant. At the cusp, μ and $\beta + \tau$ are bounded and $\text{Im } \tau \rightarrow +\infty$, so $D \rightarrow -\infty$. It is bounded above on the compact complement of a cusp neighborhood. Adding a constant with sufficiently negative imaginary part to β therefore makes $D < 0$ everywhere.

Finally,

$$\det(\operatorname{Im} Z) = 6(\operatorname{Im} \mu)^2 - \operatorname{Im} \tau \operatorname{Im} \beta = -\operatorname{Im} \tau D \neq 0.$$

Thus the four columns of Ω are real-linearly independent.

Recall from Section 5.3 that the following matrices were obtained first as the uniquely forced coordinate changes for a normalized trial period matrix, and were then defined for the final family by the specialization $b = \tau$, $d = \mu$:

$$R_{g_1} = \begin{pmatrix} -1/\tau & 0 \\ (1-\mu)/\tau & 1 \end{pmatrix}, \quad R_{g_2} = \begin{pmatrix} 1/\tau & 0 \\ -\mu/\tau & 1 \end{pmatrix}.$$

The next calculation verifies their equivariance; it does not define them again. Since $A_j^{-1} = T_j^t$, one first obtains

$$\begin{aligned} R_{g_1} \Omega &= \begin{pmatrix} -6\mu/\tau & -1 & -1/\tau & 0 \\ \beta + 6\mu(1-\mu)/\tau & 1 & (1-\mu)/\tau & 1 \end{pmatrix}, \\ R_{g_2} \Omega &= \begin{pmatrix} 6\mu/\tau & 1 & 1/\tau & 0 \\ \beta - 6\mu^2/\tau & 0 & -\mu/\tau & 1 \end{pmatrix}. \end{aligned}$$

Right multiplication by T_1^t and T_2^t , respectively, gives

$$\begin{aligned} R_{g_1} \Omega T_1^t &= \begin{pmatrix} 6(1-\mu)/\tau & (\tau-1)/\tau & 1 & 0 \\ \beta + 2 - 6(1-\mu)^2/\tau & (1-\mu)/\tau & 0 & 1 \end{pmatrix} = \Omega(g_1 z), \\ R_{g_2} \Omega T_2^t &= \begin{pmatrix} 6 + 6\mu/\tau & -1/\tau & 1 & 0 \\ \beta - 3 - 6\mu^2/\tau & 1 + \mu/\tau & 0 & 1 \end{pmatrix} = \Omega(g_2 z). \end{aligned}$$

This proves existence for the generators. Iterating each identity through its order and using $A_1^3 = A_2^4 = I$ shows that the corresponding product of the R -matrices is the identity. The presentation (4) and the cocycle rule then define R_g unambiguously for every g . This completes the construction of the proper smooth torus family over B° .

G The toric geometry needed for the cusp filling

The cusp filling uses a small, concrete part of toric geometry: lattices and their characters, smooth cones, fans and their orbit strata, and one periodic height-one fan. This appendix develops exactly that material and then translates it back into the notation of Section 6.1. No projectivity criterion, divisor-class computation, or general theory of toric degenerations is needed. The standard reference for the dictionary below is [8, Sections 1.1–1.4].

G.1 Lattices, characters, and algebraic tori

Let $\mathbf{N} \simeq \mathbb{Z}^n$ be a lattice and let

$$\mathbf{M} = \operatorname{Hom}(\mathbf{N}, \mathbb{Z})$$

be its dual, with pairing $\langle m, v \rangle$ for $m \in \mathbf{M}$ and $v \in \mathbf{N}$. The algebraic torus associated with $\mathbf{N}$ is

$$T_{\mathbf{N}} = \operatorname{Hom}(\mathbf{M}, \mathbb{C}^*) \simeq (\mathbb{C}^*)^n.$$

Each $m \in \mathbf{M}$ gives a character $\chi^m : T_{\mathbf{N}} \rightarrow \mathbb{C}^*$, and multiplication of characters records addition in the dual lattice:

$$\chi^{m+m'} = \chi^m \chi^{m'}.$$

Dually, $v \in \mathbf{N}$ gives a one-parameter subgroup $\lambda_v : \mathbb{C}^* \rightarrow T_{\mathbf{N}}$ characterized by

$$\chi^m(\lambda_v(a)) = a^{\langle m, v \rangle}.$$

Thus $\mathbf{M}$ records monomials and $\mathbf{N}$ records the directions in which those monomials acquire zeros or poles.

G.2 Cones, affine toric charts, and fans

A *strongly convex rational polyhedral cone* in $\mathbf{N}_{\mathbb{R}} = \mathbf{N} \otimes_{\mathbb{Z}} \mathbb{R}$ is a set

$$\sigma = \mathbb{R}_{\geq 0} \langle v_1, \dots, v_r \rangle, \quad v_i \in \mathbf{N}, \quad \sigma \cap (-\sigma) = \{0\}.$$

Its dual cone is

$$\sigma^{\vee} = \{m \in \mathbf{M}_{\mathbb{R}} : \langle m, v \rangle \geq 0 \text{ for every } v \in \sigma\}.$$

The semigroup $\sigma^{\vee} \cap \mathbf{M}$ determines the affine toric variety

$$U_{\sigma} = \text{Spec } \mathbb{C}[\sigma^{\vee} \cap \mathbf{M}].$$

Its dense open torus is $T_{\mathbf{N}}$, and a character χ^m extends holomorphically to U_{σ} exactly when $m \in \sigma^{\vee} \cap \mathbf{M}$.

The cone σ is *smooth* when the primitive generators of its rays extend to a basis of $\mathbf{N}$. If they are the first r members of such a basis, then

$$U_{\sigma} \simeq \mathbb{C}^r \times (\mathbb{C}^*)^{n-r}.$$

In particular, a full-dimensional cone generated by a lattice basis gives the ordinary chart $\mathbb{C}^n$. This elementary basis criterion is the source of every smoothness assertion for the fan used at the cusp.

A *fan* $\mathcal{F}$ is a collection of such cones, closed under taking faces, such that the intersection of two cones is a face of each. The affine varieties U_{σ} glue along the charts belonging to common faces and produce a toric variety $Y_{\mathcal{F}}$. When all cones are smooth, $Y_{\mathcal{F}}$ is a complex manifold. The cusp fan is infinite, but it is countable and locally finite. Local finiteness ensures that no compact region meets infinitely many unrelated charts and is the key separation property used for the analytic infinite-fan space.

G.3 Rays, divisors, and the height character

There is an inclusion-reversing correspondence

$$\{\text{cones } \sigma \in \mathcal{F}\} \longleftrightarrow \{\text{torus orbits } O(\sigma) \subset Y_{\mathcal{F}}\}, \quad \text{codim } O(\sigma) = \dim \sigma.$$

In particular, a ray ρ with primitive generator v_{ρ} determines an irreducible torus-invariant divisor D_{ρ} . For any character one has the divisor formula

$$\text{div}(\chi^m) = \sum_{\rho \in \mathcal{F}(1)} \langle m, v_{\rho} \rangle D_{\rho}, \tag{50}$$

where only rays on which the pairing is nonzero contribute locally.

Suppose now that a full-dimensional smooth cone has basis of primitive ray generators $v_0, \dots, v_{n-1}$, with dual basis $m_0, \dots, m_{n-1}$. The associated affine coordinates are $z_i = \chi^{m_i}$. If a character $h \in \mathbf{M}$ satisfies $h(v_i) = 1$ for every ray generator, then

$$h = m_0 + \dots + m_{n-1}, \quad \chi^h = z_0 \cdots z_{n-1}.$$

Consequently the fibre ($\chi^h = 0$) is reduced, and in this chart it is the union of the coordinate hyperplanes. If the same height-one condition holds on every ray of a smooth fan, the central fibre is therefore a reduced simple-normal-crossing divisor. This is the entire toric reason for the local equation $t_c = z_0 z_1 z_2$ in complex dimension three.

The toric variety of D_ρ is read from the *star* of ρ : project the cones containing ρ to the quotient lattice $\mathbf{N}/\mathbb{Z}v_\rho$. The resulting fan describes the normalization and boundary strata of that divisor. We shall use this once, to identify each component upstairs as dP_6 .

G.4 The periodic height-one fan used here

Take

$$\mathbf{N}' = \mathbb{Z}^2 \oplus \mathbb{Z},$$

and let $h \in (\mathbf{N}')^\vee$ be projection to the last coordinate. Triangulate the height-one plane by

$$\begin{aligned} P_v^+ &= \text{conv}\{v, v + e_1, v + e_2\}, \\ P_v^- &= \text{conv}\{v + e_1, v + e_2, v + e_1 + e_2\}, \quad v \in \mathbb{Z}^2. \end{aligned}$$

For every triangle take the cone from the origin over $P_v^\pm \times \{1\}$, together with all faces. This is the fan $\mathcal{F}$ used in the cusp model.

The basic triangle P_0^+ has ray generators

$$v_0 = (0, 0, 1), \quad v_1 = (1, 0, 1), \quad v_2 = (0, 1, 1).$$

They form a basis of $\mathbf{N}'$, with dual basis

$$m_0 = (-1, -1, 1), \quad m_1 = (1, 0, 0), \quad m_2 = (0, 1, 0).$$

On the dense torus write

$$x_1 = \chi^{(1,0,0)}, \quad x_2 = \chi^{(0,1,0)}, \quad t_c = \chi^{(0,0,1)}.$$

The affine coordinates of this cone are therefore

$$(z_0, z_1, z_2) = \left(\frac{t_c}{x_1 x_2}, x_1, x_2 \right), \quad t_c = z_0 z_1 z_2. \quad (51)$$

The other triangles give the same determinant-one calculation after an integral change of basis. Hence the entire infinite fan is smooth, and because every ray lies at height one, its central divisor is reduced with simple normal crossings.

Fix a vertex of the planar triangulation. After quotienting by the corresponding height-one ray, the six adjacent edges give the cyclic ray directions

$$e_1, e_2, e_2 - e_1, -e_1, -e_2, e_1 - e_2.$$

This is the hexagonal fan of the toric del Pezzo surface dP_6 , equivalently the blow-up of $\mathbb{P}^2$ at its three torus-fixed points. Its toric boundary is a hexagon of six (-1) -curves. Thus every component of the unquotiented central divisor is a copy of dP_6 .

G.5 Why the infinite fan has a finite quotient

For $a \in \mathbb{Z}^2$, the map

$$\Phi_a : \mathbf{N}' \longrightarrow \mathbf{N}', \quad \Phi_a(n, k) = (n + ka, k),$$

is an integral linear automorphism preserving the height character. On the height-one plane it is the translation $n \mapsto n + a$, so it preserves the periodic fan and induces a toric automorphism of $Y_{\mathcal{F}}$. On the dense torus this automorphism is represented by

$$(x, t_c) \longmapsto (t_c^a x, t_c),$$

where powers and products in the two x -coordinates are taken componentwise. Multiplication by a holomorphic unit $c_a(t_c) \in (\mathbb{C}^*)^2$ is an ordinary torus translation and extends as well. This explains the two factors in the action

$$x \longmapsto c_{\bar{\lambda}}(t_c) t_c^{B_0 \bar{\lambda}} x$$

from (25): the monomial factor shifts the fan, while the unit factor translates inside its torus orbits.

Modulo the full translation lattice $\mathbb{Z}^2$, the planar triangulation has one vertex orbit, three edge orbits, and two triangle orbits. Accordingly, the central quotient has normalization dP_6 , three double curves, and two triple points. In particular, the infinite collection of components upstairs becomes one irreducible compact component downstairs. This finite cell-orbit calculation explains compactness of the central fibre; it does *not* by itself prove properness of the total family over a disc. That stronger analytic statement requires the uniform logarithmic fundamental-domain estimate in Appendix H.

G.6 Dictionary for the cusp construction

The following table matches the abstract toric language with the objects already constructed in the main text.

Object from Section 6.1	Toric role
Λ_{tor}	Dividing by its two invariant periods and exponentiating changes the additive fibre $\mathbb{C}^2$ into the dense torus $(\mathbb{C}^*)^2$.
$\bar{\Lambda} = \Lambda / \Lambda_{\text{tor}}$	The residual rank-two lattice that must still be divided out.
$B_0 : \bar{\Lambda} \rightarrow \mathbb{Z}^2$	Identifies the residual lattice with the full translation lattice of the periodic fan; unimodularity is what makes the cell quotient have the orbit counts above.
$Z = sB_0 + C(t_c)$	After exponentiation, the singular term becomes the monomial $t_c^{B_0 \bar{\lambda}}$ and the holomorphic term becomes the unit $c_{\bar{\lambda}}(t_c)$.
Height-one fan $\mathcal{F}$	Supplies smooth three-dimensional toric charts in which the map is $t_c = z_0 z_1 z_2$, hence a reduced simple-normal-crossing central fibre.
Periodic fan quotient	Turns infinitely many dP_6 components upstairs into the compact surface W with one normalized component, three double curves, and two triple points.

With this dictionary in place, the remaining claims in the cusp construction are analytic rather than formal toric facts: the residual action must be free and properly discontinuous, the quotient must be Hausdorff, and the map to the disc must be proper. These are proved in the next appendix.

H The cusp quotient in local coordinates

We give the full fan and the quotient estimates summarized in Section 6.1. Recall from (7) that the toric sublattice is

$$\Lambda_{\text{tor}} = \mathbb{Z}\langle \widehat{w}, \widehat{\delta} \rangle.$$

Its torsion-free quotient and the ambient toric lattice are

$$Q := \Lambda / \Lambda_{\text{tor}} \simeq \mathbb{Z}^2, \quad N' = \mathbb{Z}^2 \oplus \mathbb{Z}.$$

Triangulate the height-one plane with vertex set $\mathbb{Z}^2$ and triangles

$$\text{conv}\{v, v + e_1, v + e_2\}, \quad \text{conv}\{v + e_1, v + e_2, v + e_1 + e_2\}.$$

For every cell P , take the cone $\widehat{P} = \text{cone}(P \times \{1\}) \subset N'_{\mathbb{R}}$. These cones and their faces form a locally finite fan $\mathcal{F}$. The primitive generators of each maximal cone have determinant ± 1 : subtracting one height-one generator from the other two reduces the determinant to that of e_1, e_2 . Hence every affine toric chart is $\mathbb{C}^3$ and $Y = Y_{\mathcal{F}}$ is smooth. The common-face condition gives separation; local finiteness reduces any two-point separation problem to a finite subfan. Thus the infinite fan still defines a Hausdorff, second-countable complex threefold.

The height character pairs to 1 with every ray generator. In a maximal chart, if m_0, m_1, m_2 is the dual basis, the height is $m_0 + m_1 + m_2$, whence $t_c = z_0 z_1 z_2$. The star of a ray has cyclic rays

$$e_1, e_2, e_2 - e_1, -e_1, -e_2, e_1 - e_2,$$

the hexagonal fan of dP_6 . Therefore every irreducible component of the unquotiented central divisor is a copy of dP_6 .

Translation of the planar triangulation by $B_0 Q = \mathbb{Z}^2$ preserves the fan. The bounded factor $c_{\bar{\lambda}}(t_c)$ in (25) is a torus translation, so the formula extends from the dense torus to a biholomorphism of Y ; additivity makes these maps an action of Q .

For $0 < |t_c| < \epsilon$, put $y = \log |x| / \log |t_c|$. The action becomes

$$y \mapsto y + B_{t_c} \bar{\lambda}, \quad B_{t_c} = B_0 + \frac{-2\pi \text{Im } C(t_c)}{\log |t_c|}.$$

Because C is bounded and B_0 is invertible, B_{t_c} and its inverse are uniformly bounded for small $|t_c|$. The parallelograms $B_{t_c}([-1/2, 1/2]^2)$ therefore lie in one fixed compact polygon and meet only finitely many cells of the locally finite fan. Taking the corresponding closed toric chart regions and the compact angular torus produces a compact set meeting every orbit, uniformly as $t_c \rightarrow 0$. This proves properness of the map from the quotient to the disc.

To prove proper discontinuity, let K_1, K_2 be compact subsets of the restricted toric family. Their logarithmic coordinates meet only finitely many cells and lie in a bounded region. If $\bar{\lambda} K_1 \cap K_2 \neq \emptyset$, uniform invertibility of B_{t_c} bounds $\bar{\lambda}$. Only finitely many lattice vectors can occur. At $t_c = 0$, the same conclusion follows because a fan translation must carry one of finitely many cells meeting K_1 to one meeting K_2 .

For freeness on the punctured family, a fixed point would imply

$$|\log |t_c|| |B_0 \bar{\lambda}| \leq |R(t_c) \bar{\lambda}|.$$

The right side is bounded by a constant times $|\bar{\lambda}|$, whereas unimodularity gives $|B_0 \bar{\lambda}| \geq c |\bar{\lambda}|$ for a fixed $c > 0$. The inequality is impossible for sufficiently small $|t_c|$ and $\bar{\lambda} \neq 0$. On the

central fibre a fixed point would require a nonzero fan translation to stabilize a cone, but translation of a bounded cell cannot fix that cell. The action is therefore free everywhere.

The free properly discontinuous quotient theorem now gives a Hausdorff complex manifold N_0 . Since B_0 acts transitively on vertices, the normalization of the central fibre has one component, dP_6 . Opposite boundary edges are identified, producing three double curves. The triangulation has two orbits of oriented triangles, producing two triple points. These facts also justify the stratification used in the nearby-cycle calculation.

H.1 Agreement with the punctured torus family

We finally verify that the toric quotient fills the specific family J , not merely a fibrewise isomorphic one. The translations by Λ_{tor} disappear under the first two exponentials in E_{exp} . Moreover, since t_c is g_0 -invariant and $s \circ g_0 = s - 1$, formula (22) gives

$$Z(g_0 z) = Z(z) - B_0.$$

In the ordered basis $(\hat{\gamma}, \hat{u}, \hat{w}, \hat{\delta})$, the inverse cusp action is

$$A_0^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

Multiplication of the four period columns by this matrix shows that the preceding identity is equivalent to

$$\Omega(g_0 z) = \Omega(z) A_0^{-1}.$$

Thus the complex-linear coordinate change in (21) is $R_{g_0} = I_2$. Together with $e^{2\pi i(s-1)} = e^{2\pi i s}$, this proves that E_{exp} is invariant under the cusp deck transformation.

For a general $\lambda \in \Lambda$, the first two exponentials change by the multiplier in (23). This is exactly the action $\Psi_{\bar{\lambda}}$, where $\bar{\lambda}$ is the image of λ in Q . Hence E_{exp} descends to the biholomorphism (27). On the cone over $\text{conv}\{0, e_1, e_2\}$ the affine toric coordinates are

$$(u_0, u_1, u_2) = \left(\frac{t_c}{x_1 x_2}, x_1, x_2 \right),$$

so the zero section becomes $(t_c, 1, 1)$. It extends across the origin and meets the smooth stratum of the central fibre transversely.

I The elliptic quotients and compact gluing

I.1 The affine fixed-point criterion

Let $A \in \text{GL}(\Lambda)$ have finite order and $b \in \Lambda \otimes \mathbb{Q}$. The affine map $x \mapsto Ax + b$ on $\Lambda_{\mathbb{R}}/\Lambda$ has a fixed point if and only if

$$\theta(b) \in \mathbb{Z} \quad \text{for every } \theta \in V^A = \{\theta \in \text{Hom}(\Lambda, \mathbb{Z}) : \theta A = \theta\}. \quad (52)$$

Indeed, the fixed-point equation is $b \in (I - A)\Lambda_{\mathbb{R}} + \Lambda$. Since A has finite order, $(I - A)\Lambda_{\mathbb{R}}$ is rational, and

$$K = \Lambda_{\mathbb{R}} / ((I - A)\Lambda_{\mathbb{R}} + \Lambda)$$

is a compact torus. Its integral characters are precisely the invariant functionals V^A . Characters separate points, so the class of b in K vanishes exactly when all of them take integral values on b .

Apply this to a nonidentity power of $\tilde{g}_j$. Such a power can fix only the central torus, where its linear and translation parts are A_j^k and kv_j/m_j . The invariant functional γ takes the values $k/3$ for $j = 1$ and $-k/4$ for $j = 2$, never an integer in the relevant ranges. Criterion (52) proves freeness of the two cyclic actions. Finite quotients preserve properness, so the local maps $N_j \rightarrow D_j$ are proper smooth complex threefolds.

I.2 The punctured logarithmic identification

On a sector of the punctured s_j -disc choose a branch of $\log s_j$ and put

$$\sigma_j(z) = \frac{\log s_j(z)}{2\pi i} \Omega(z) v_j.$$

The identities $A_j v_j = v_j$ and $s_j(g_j z) = e^{-2\pi i/m_j} s_j(z)$ give

$$\sigma_j(g_j z) = R_{g_j}(z) \sigma_j(z) - \Omega(g_j z) \frac{v_j}{m_j}.$$

Hence translation by σ_j conjugates the affine cyclic action to the untwisted monodromy action. Changing the logarithm by $2\pi i n$ changes σ_j by the integral period $n\Omega(z)v_j$, so the translation is single valued on the torus quotient. This proves the biholomorphism (29) and shows that the quotient is a genuine logarithmic transform of the original punctured family.

I.3 Global separation and compactness

Choose the three discs with pairwise disjoint closures. Each overlap with the pair-of-pants core is an annulus, and the preceding identifications are biholomorphisms over those annuli. The gluing relation is therefore an equivalence relation generated by homeomorphisms between open subsets. Two points over different base values are separated downstairs; two points over one base value lie in a single Hausdorff local model. Thus the glued space is Hausdorff. Properness is local on the target: over each disc or a relatively compact subset of the core, the inverse image of a compact set is compact. A finite cover of the compact base $\mathbb{P}^1$ then shows that X is compact. The charts glue holomorphically, so X is a compact complex threefold before any topological recognition is invoked.

J Van Kampen and the exact fundamental group

Over the pair-of-pants core, the fundamental group is the semidirect product described explicitly in (30). The logarithmic cap over p_j contributes the relation $\alpha_j^{m_j} = t_{v_j}$: after m_j turns the base coordinate closes and the accumulated fractional translations add to v_j . The cusp cap kills Λ_{tor} and the meridian $(\alpha_1 \alpha_2)^{-1}$.

Taking the normal closure under the two monodromies kills all of $L = \ker \gamma$ by Appendix E. The fibre group therefore reduces to the central infinite cyclic group generated by the class c of $\hat{\gamma}$. If the cusp gluing is displaced by ℓ_0 in this quotient, put $x = \alpha_1$ and $y = \alpha_2$. Van Kampen gives precisely (33).

Eliminating $y = x^{-1}c^{\ell_0}$ makes the group abelian with relation matrix

$$R = \begin{pmatrix} 3 & -\ell_1 \\ 4 & \ell_2 - 4\ell_0 \end{pmatrix}.$$

The greatest common divisor of all entries is one because 3 and 4 occur. The first Smith invariant is therefore one, while

$$\det R = 3\ell_2 - 12\ell_0 + 4\ell_1 = -p.$$

Thus the group is cyclic of order $|p|$. For $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$, $p = -1$, proving $\pi_1(X) = 1$.

K Integral nearby cycles at the cusp

We now justify the specialization isomorphism (40), including its integral saturation.

K.1 The two resolutions on the central fibre

The local equation at a point where r components meet is $z_0 \cdots z_{r-1} = t_c$. Its Milnor fibre deformation retracts onto $(S^1)^{r-1}$. The only nearby cohomology sheaves are therefore those listed before (35). Along a double curve, the generator is the oriented phase difference. Toric translations multiply the two coordinates by reciprocal units, so this orientation has no sign monodromy.

At each triple point, the three oriented circle classes obey $n_1 - n_2 + n_3 = 0$. The two triple points receive the same signed sum after choosing the compatible orientations induced by the two triangle orbits. This proves the exact resolution (35). Its cohomology is computed from the map

$$H^0(D_1 \sqcup D_2 \sqcup D_3; \mathbb{Z}) = \mathbb{Z}^3 \longrightarrow \mathbb{Z}^2, \quad (n_1, n_2, n_3) \longmapsto (s, s).$$

The image is the diagonal copy of $\mathbb{Z}$, the kernel has rank two, and the cokernel has rank one. Including the degree-two cohomology of the three $D_k \simeq \mathbb{P}^1$ gives

$$H^\bullet(W; \mathcal{V}_{\text{van}}) = (\mathbb{Z}^2, \mathbb{Z}, \mathbb{Z}^3, 0).$$

For the constant sheaf, the normalization sequence is (36). Since $H^\bullet(dP_6) = (\mathbb{Z}, 0, \mathbb{Z}^4, 0, \mathbb{Z})$ and each double curve has cohomology $(\mathbb{Z}, 0, \mathbb{Z})$, the long exact sequence gives

$$H^\bullet(W; \mathbb{Z}) = (\mathbb{Z}, \mathbb{Z}^2, \mathbb{Z}^4, \mathbb{Z}^2, \mathbb{Z}).$$

Substitution into the nearby-cycle spectral sequence produces the rank table displayed after (37). Comparison with the Betti numbers $(1, 4, 6, 4, 1)$ of T^4 determines every differential except a possible integral ambiguity in total degree three.

K.2 The primitive torus and saturation

Choose a compact real torus K in the dense torus of W_{reg} . Under the exponential gluing, a nearby translate K_t has lattice

$$\pi_1(K_t) = \mathbb{Z}\langle \hat{w}, \hat{\delta} \rangle.$$

This is a primitive direct summand of Λ , with complementary summand $\mathbb{Z}\langle \hat{\gamma}, \hat{u} \rangle$. Hence the nearby fibre splits topologically as $T_f \times T_B$, with K_t a translate of T_f .

Excision and Thom isomorphism identify the vertical maps in the square in Section 9. The lower horizontal map is

$$H^1(T_f) \longrightarrow H^3(T_f \times T_B), \quad a \longmapsto a \times [T_B]^\vee.$$

By the integral Künneth theorem, its image is a primitive rank-two direct summand. The six boundary curves of dP_6 show that the supported classes upstairs generate $H^3(W) \simeq \mathbb{Z}^2$. Therefore specialization in degree three is injective and its image is exactly that primitive Gysin image.

The remaining filtration quotients of the nearby-cycle spectral sequence are free: in degree one the quotient is $\mathbb{Z}^2$; in degree two the pieces are $\mathbb{Z}$ and a kernel between free groups; degrees zero and four are immediate. Thus specialization is injective with saturated image in every degree. Its image is contained in the kernel of $T_0 - I$, itself a saturated lattice. Both lattices have ranks $(1, 2, 4, 2, 1)$ in degrees 0 through 4. Equal-rank saturated inclusions have trivial finite quotient, proving (40) integrally.

L Elliptic specialization lattices and cohomology on the base

Define

$$\eta_1 = 2u + w + 3\delta, \quad \eta_2 = u + w + 2\delta.$$

Integer row reduction gives

$$V^{T_j} = \mathbb{Z}\langle \gamma, \eta_j \rangle, \quad \left(\bigwedge^2 V \right)^{T_j} = \mathbb{Z}\langle \gamma \wedge \eta_j, q_0 \rangle.$$

Let $\pi_j : F \rightarrow S_j$ be the free cyclic cover of degree m_j . The abelianization of

$$\langle \Lambda, \tilde{g}_j \mid \tilde{g}_j \lambda \tilde{g}_j^{-1} = A_j \lambda, \tilde{g}_j^{m_j} = t_{v_j} \rangle$$

is torsion free of rank two, because the relation $([v_j], -m_j)$ is primitive: its γ -coordinate is ± 1 . Since $e(S_j) = e(F)/m_j = 0$, Poincaré duality gives torsion-free cohomology of ranks $(1, 2, 2, 2, 1)$. Transfer makes every pullback π_j^* injective.

The transgression in the cyclic group-extension spectral sequence is evaluation on $[v_j]$. Therefore

$$\pi_j^* H^1(S_j) = \{ \theta \in V^{T_j} : \theta(v_j) \equiv 0 \pmod{m_j} \}.$$

Since $\eta_j(v_j) = 0$, this yields

$$\pi_1^* H^1(S_1) = \mathbb{Z}\langle 3\gamma, \eta_1 \rangle, \quad \pi_2^* H^1(S_2) = \mathbb{Z}\langle 4\gamma, \eta_2 \rangle.$$

In the displayed invariant bases, the degree-two cup-product matrices are

$$\begin{pmatrix} 0 & 3 \\ 3 & 12 \end{pmatrix}, \quad \begin{pmatrix} 0 & 2 \\ 2 & 12 \end{pmatrix}.$$

If k_j is the index of $\pi_j^* H^2(S_j)$, the projection formula and unimodularity of the intersection form give $9k_1^2 = 3^2$ and $4k_2^2 = 4^2$. Hence $k_1 = 1$ and $k_2 = 2$. At p_2 , contraction with v_2 gives

$$d_2(\gamma \wedge \eta_2) = -[\eta_2] \neq 0 \quad \text{in} \quad \frac{V^{T_2}}{(I + T_2 + T_2^2 + T_2^3)V} = \frac{\mathbb{Z}\langle \gamma, \eta_2 \rangle}{\mathbb{Z}\langle 4\gamma, 2\eta_2 \rangle}.$$

Thus the index-two lattice cannot contain $\gamma \wedge \eta_2$ as a primitive generator. Moreover K_{S_2} is torsion because its pullback to F is trivial. As $H^2(S_2; \mathbb{Z})$ is torsion free, $w_2(S_2) = 0$, so its intersection form is even. These two conditions select

$$\pi_2^* H^2(S_2) = \{a\gamma \wedge \eta_2 + bq_0 : a \equiv b \pmod{2}\}.$$

Perfect pairings between degrees one and three give the degree-three indices, and the orientation class pulls back with degree m_j . This proves the table (41) and identifies all its generators.

Let $j : B^\circ \hookrightarrow B$. The cusp calculation gives no quotient between $R^q f_* \mathbb{Z}$ and $j_* L^q$ at p_0 . The preceding index computation gives skyscraper quotients

q	$j_* L^q / R^q f_* \mathbb{Z}$
1	$(\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2}$
2	$(\mathbb{Z}/2)_{p_2}$
3	$(\mathbb{Z}/2)_{p_2}$
4	$(\mathbb{Z}/3)_{p_1} \oplus (\mathbb{Z}/4)_{p_2}$.

Intersecting the global invariant lines with these local lattices gives (43). Poincaré duality with local coefficients on the punctured sphere gives $H^2(B, R^q f_* \mathbb{Z}) \simeq (\wedge^q V)_G$.

For completeness, a one-cocycle on B° is determined by its values on meridians a_1, a_2 . Extension to $j_* L^q$ requires the value at each meridian to lie in $(T_i - I)M$. Subtract a coboundary to make the value on a_2 zero. The remaining group is

$$H^1(B, j_* \mathcal{L}) = \frac{(T_1 - I)M \cap (T_0 - I)M}{(T_1 - I)M^{T_2}}.$$

For $M = V$, numerator and denominator are both $\mathbb{Z}(2\gamma - u)$; for $M = \wedge^2 V$, both are $\mathbb{Z}(\gamma \wedge u)$. Thus both groups vanish. Finally, global invariants surject onto the skyscraper quotients: γ maps to $(1, 1)$ in $\mathbb{Z}/3 \oplus \mathbb{Z}/4$, and q_0 maps to the nonzero element of $\mathbb{Z}/2$. The long exact sequence therefore proves (45).

M The three integral Leray transgressions

Cover B by a pair of pants P and cap discs D_0, D_1, D_2 . Orient the annuli so that

$$[\partial P] = [\partial D_0] - [\partial D_1] - [\partial D_2].$$

If local integral cocycles representing a section of $R^1 f_* \mathbb{Z}$ differ on the three annuli by integers n_0, n_1, n_2 , the Mayer–Vietoris double complex filtered by base degree identifies the Leray differential with the boundary map. Evaluation on $[B]$ gives

$$d_2(a) = (n_0 - n_1 - n_2)\eta.$$

Changing any cocycle representative changes the annular triple by an element in the kernel of the Mayer–Vietoris boundary, so the formula is choice independent and integral.

For $a = 12\gamma$, define the cocycle on the core by its value 12γ on Λ and zero on the base generators. On the j th logarithmic cap, set

$$\chi_j|_\Lambda = 12\gamma, \quad \chi_j(\tilde{g}_j) = \frac{12}{m_j} \gamma(v_j).$$

This is integral and respects both the conjugation and power relations. On the cusp cap, the corresponding value is $12\ell_0$. Hence the three annular differences are $12\ell_0, 4\ell_1, 3\ell_2$, and the oriented boundary formula proves (46).

Write the next coefficients as p', p'' as in Section 10. The product

$$(12\gamma)(2q_0) = 12(2\gamma \wedge u \wedge w)$$

and the relation $[q_0] = 12[\gamma \wedge \delta]$ give, by the signed derivation rule, $p'' = 2p - p'$. The product $(2q_0)(2\gamma \wedge u \wedge w)$ vanishes for degree reasons. Its differential uses the primitive pairings

$$[\delta](2\gamma \wedge u \wedge w) = -2[\text{vol}], \quad (2q_0)[\gamma \wedge \delta] = 2[\text{vol}].$$

Thus $p' = p''$. Combining the two equations gives $p' = p'' = p$, proving (47) without a rational ambiguity.

When $p = -1$, the three maps are isomorphisms of primitive infinite cyclic groups. Together with the vanishing H^1 rows, the Leray filtration gives (48). The universal coefficient and Poincaré-duality argument in Section 11 then gives the integral homology sphere, and the classical recognition theorems in Appendix D finish the identification with the standard smooth S^6 .

N Notation and terminology

Toric sublattice Λ_{tor}

The primitive rank-two sublattice $\ker(A_0 - I) = \text{im}(A_0 - I) = \mathbb{Z}\langle \hat{w}, \hat{\delta} \rangle$ of the fibre lattice Λ . Here tor abbreviates its toric or toroidal role, not torsion. Dividing by its periods and exponentiating gives $(\mathbb{C}^*)^2$; the quotient $\Lambda/\Lambda_{\text{tor}} \simeq \mathbb{Z}^2$ is torsion free.

Bielliptic surface

A compact complex surface $(E \times F)/G$ obtained from elliptic curves and a finite free action, with Kodaira dimension zero and irregularity one.

Logarithmic transform

A surgery on a torus fibration obtained by quotienting a pulled-back family by a base rotation combined with a fibre translation.

Normal-crossing fibre

A divisor locally defined by a product of coordinate functions. The fibre W is irreducible globally but non-normal; its normalization is dP_6 .

Saturated sublattice

A sublattice $L \subset M$ for which M/L is torsion free. Equal rank over $\mathbb{Q}$ does not imply equality over $\mathbb{Z}$; saturation is what removes the finite-index ambiguity.

Nearby cycles

The sheaf-theoretic object recording the cohomology of local Milnor fibres as a singular fibre is approached.

Algebraic dimension

The transcendence degree of the field of meromorphic functions. It is not needed for the recognition of S^6 and is therefore not developed here.

Source note

The principal source is the 108-page anonymous manuscript *A compact complex threefold fibred by tori over the projective line, and the six-sphere*, together with its two-page synopsis titled *The $(3, 4, \infty)$ modular family of 2-tori, completed at its three special points, is a complex structure on S^6* . The version consulted was downloaded from <https://alpo.ge/s6.pdf> on 28 August 2026; its SHA-256 digest is

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